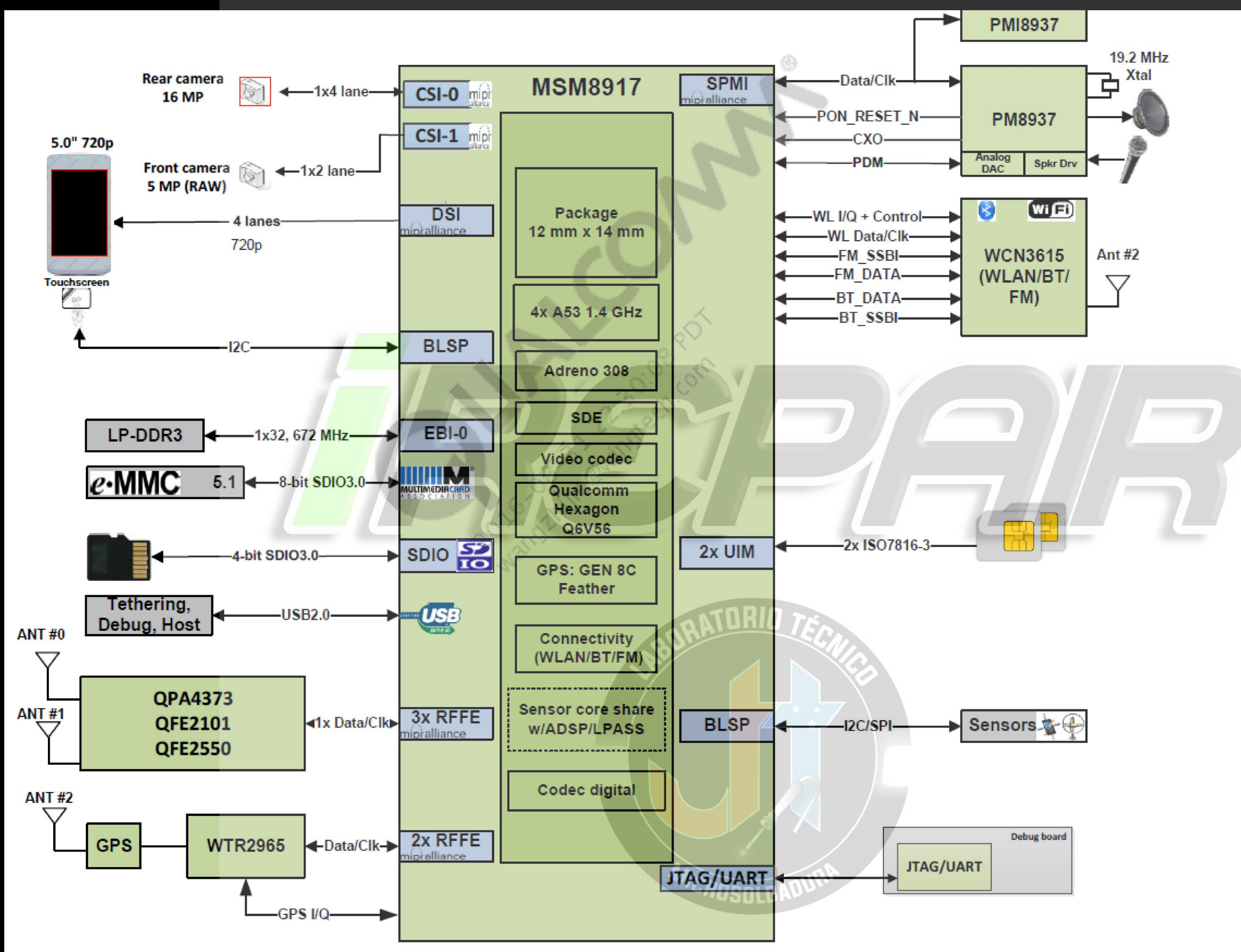




Block Diagram

MSM8937 GPIO Configuration					
GPIO_0	NC	GPIO_41	SCAM_ID	GPIO_82	FM_DATA
GPIO_1	NC	GPIO_42	ACCL_INT1_N	GPIO_83	BT_CTL
GPIO_2	NC	GPIO_43	ALSP_INT_N	GPIO_84	BT_DATA
GPIO_3	WDOG_DISABLE	GPIO_44	MAG_RST_N	GPIO_85	NC
GPIO_4	UART_MSM_TX	GPIO_45	NC	GPIO_86	NC
GPIO_5	UART_MSM_RX	GPIO_46	NC	GPIO_87	NC
GPIO_6	NC	GPIO_47	NC	GPIO_88	NC
GPIO_7	NC	GPIO_48	NC	GPIO_89	NC
GPIO_8	NC	GPIO_49	UIM_BATT_ALM	GPIO_90	NC
GPIO_9	NC	GPIO_50	NC	GPIO_91	KEY_VOL_UP_N
GPIO_10	TP_I2C_SDA	GPIO_51	UIM1_DATA	GPIO_92	NC
GPIO_11	TP_I2C_SCL	GPIO_52	UIM1_CLK	GPIO_93	NC
GPIO_12	NC	GPIO_53	UIM1_RESET	GPIO_94	NC
GPIO_13	NC	GPIO_54	NC	GPIO_95	NC
GPIO_14	SENSOR_I2C_SDA	GPIO_55	UIM2_DATA	GPIO_96	NC
GPIO_15	SENSOR_I2C_SCL	GPIO_56	UIM2_CLK	GPIO_97	NC
GPIO_16	NC	GPIO_57	UIM2_RESET	GPIO_98	NC
GPIO_17	NC	GPIO_58	NC	GPIO_99	NC
GPIO_18	NC	GPIO_59	NC	GPIO_100	RFFE1_CLK
GPIO_19	NC	GPIO_60	LCD_RST_N	GPIO_101	RFFE1_DATA
GPIO_20	ACC_SPI_MOSI	GPIO_61	LCD_ID	GPIO_102	NC
GPIO_21	ACC_SPI_MISO	GPIO_62	NC	GPIO_103	NC
GPIO_22	ACC_SPI_CS	GPIO_63	APP_BOOT_FROM_ROM	GPIO_104	NC
GPIO_23	ACC_SPI_CLK	GPIO_64	TP_RST_N	GPIO_105	GRFC1_SEL
GPIO_24	LCD_TE0	GPIO_65	TP_INT_N	GPIO_106	NC
GPIO_25	NC	GPIO_66	GP_PDM_A0	GPIO_107	GRFC3_SEL
GPIO_26	CAM_MCLK0	GPIO_67	SDCARD_DET_N	GPIO_108	NC
GPIO_27	NC	GPIO_68	NC	GPIO_109	GRFC5_SEL
GPIO_28	CAM_MCLK2	GPIO_69	CDC_PDM_CLK	GPIO_110	GRFC6_SEL
GPIO_29	CAM_I2C_SDA0	GPIO_70	CDC_PDM_SYNC	GPIO_111	GRFC7_SEL
GPIO_30	CAM_I2C_SCL0	GPIO_71	CDC_PDM_TX	GPIO_112	GRFC8_SEL
GPIO_31	NC	GPIO_72	CDC_PDM_RX0	GPIO_113	GRFC9_SEL
GPIO_32	NC	GPIO_73	CDC_PDM_RX1	GPIO_114	GRFC10_SEL
GPIO_33	NC	GPIO_74	CDC_PDM_RX2	GPIO_115	GRFC11_SEL
GPIO_34	FLASH_STROBE_NOW	GPIO_75	BT_SSBI	GPIO_116	GRFC12_SEL
GPIO_35	MCAM1_PWDN	GPIO_76	WL_CMD_DATA_2	GPIO_117	NC
GPIO_36	NC	GPIO_77	WL_CMD_DATA_1	GPIO_118	EXT_GPS_LNA_EN
GPIO_37	FORCE_USB_BOOT	GPIO_78	WL_CMD_DATA_0	GPIO_119	CH0_GSM_TX_PHASE_D0
GPIO_38	NC	GPIO_79	WL_CMD_SET	GPIO_120	RFFE5_CLK
GPIO_39	SCAM_PWDN	GPIO_80	WL_CMD_CLK	GPIO_121	RFFE5_DATA
GPIO_40	SCAM_RST_N	GPIO_81	FM_SSBI	GPIO_122	NC

GPIO_123	NC
GPIO_124	NC
GPIO_125	NC
GPIO_126	NC
GPIO_127	NC
GPIO_128	NC
GPIO_129	NC
GPIO_130	NC
GPIO_131	NC
GPIO_132	NC
GPIO_133	NC

PMI8937 MPP Configuration			
MPP_1	NC		
MPP_2	GREEN_LED		
MPP_4	FLASH_STROBE_NOW		

PM8937 GPIO/MPP Configuration			
GPIO_1	NC	MPP_1	VDD_PX_BIAS_MPP_1
GPIO_2	SDCARD_DET_N	MPP_2	NC
GPIO_3	UIM_BATT_ALM	MPP_3	VREF_DAC_MPP_3
GPIO_4	NC	MPP_4	QUIET_THERM
GPIO_5	NC		
GPIO_6	NC		
GPIO_7	NC		
GPIO_8	NC		

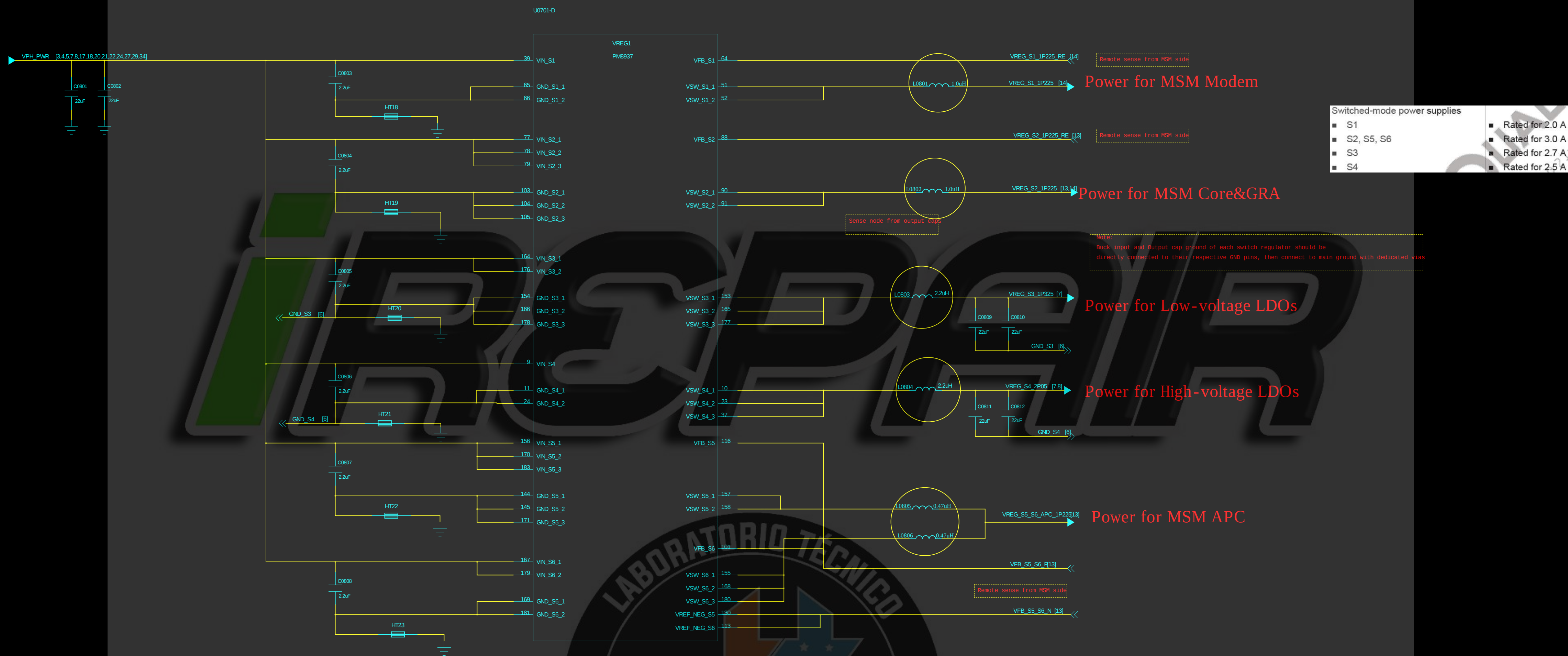
GPIO MAP

PMI8940 CHARGER

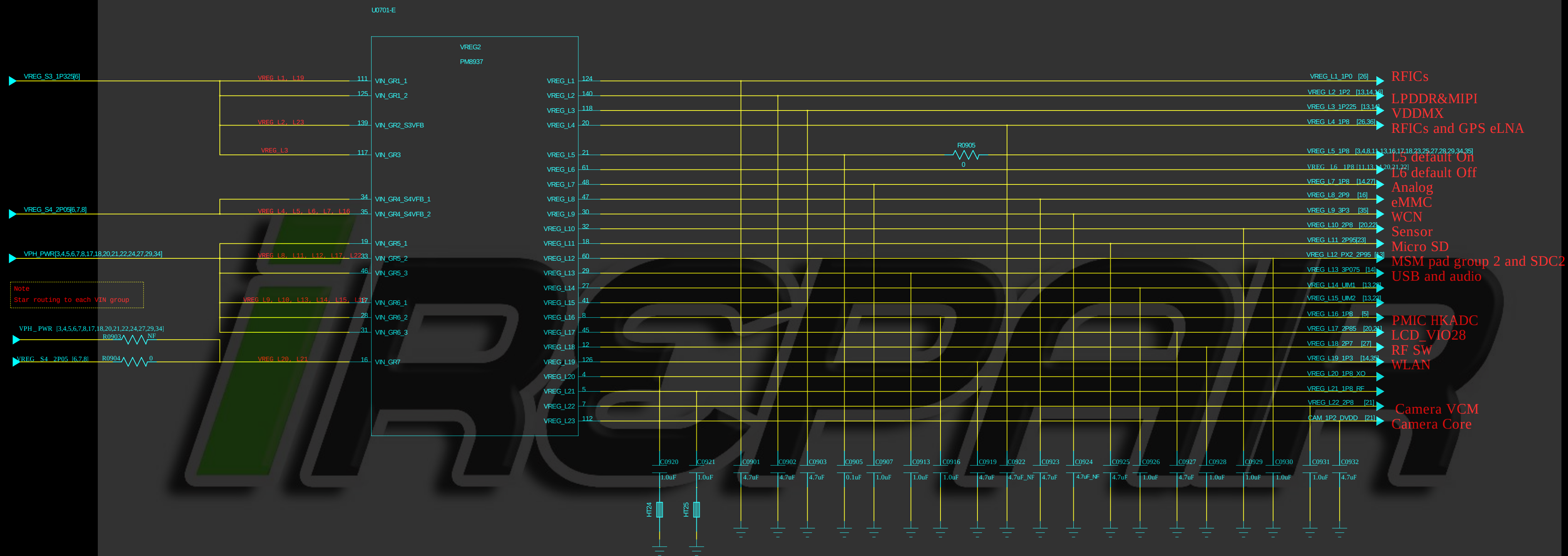
BATTERY CONNECTOR

PMI8940 Control/Interface

PM8937 BUCK CONVERTER

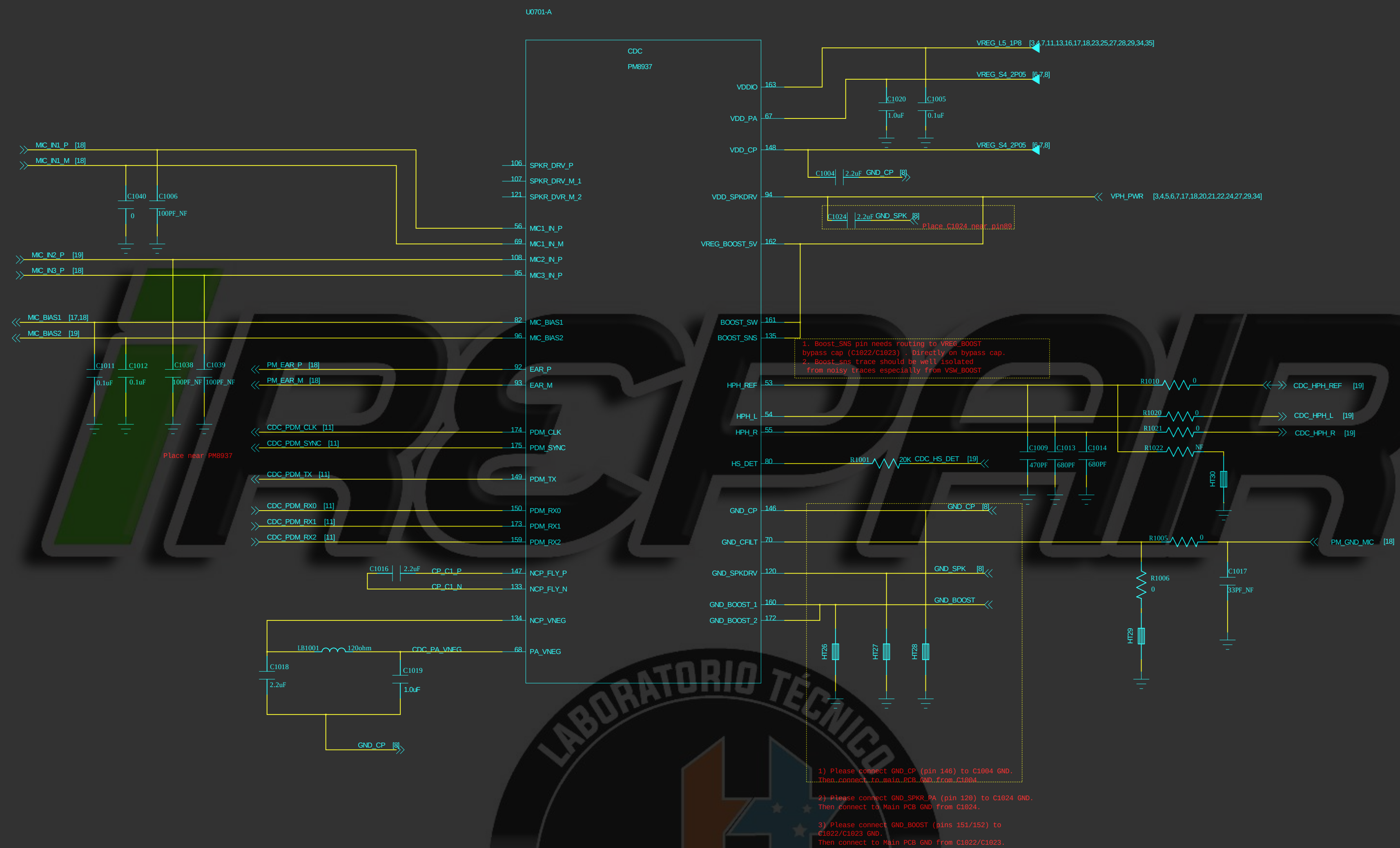


PM8937 LDO CIRCUITS



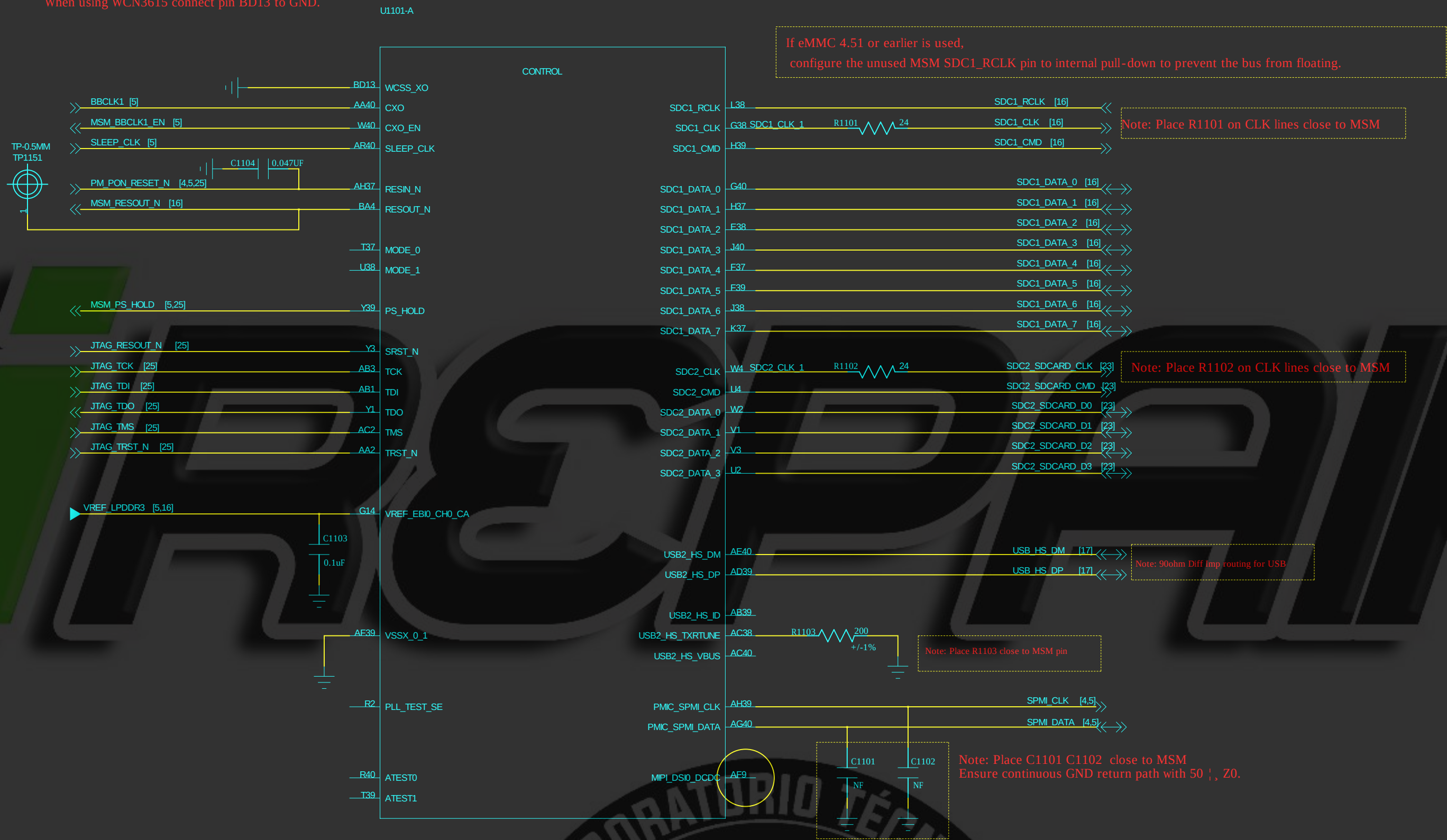
PSEUDO CAPLESS LDOs
L4*, L6, L8*, L9*, L10, L11*, L12, L14, L15, L17*, L18, L22, L23
L4*, L8*, L9*, L11*, L17*, Still need validation as Pseudo-capless
L5, L7, L13, L16 have internal PMIC loads and require local CAPS.

PM8937 Codec



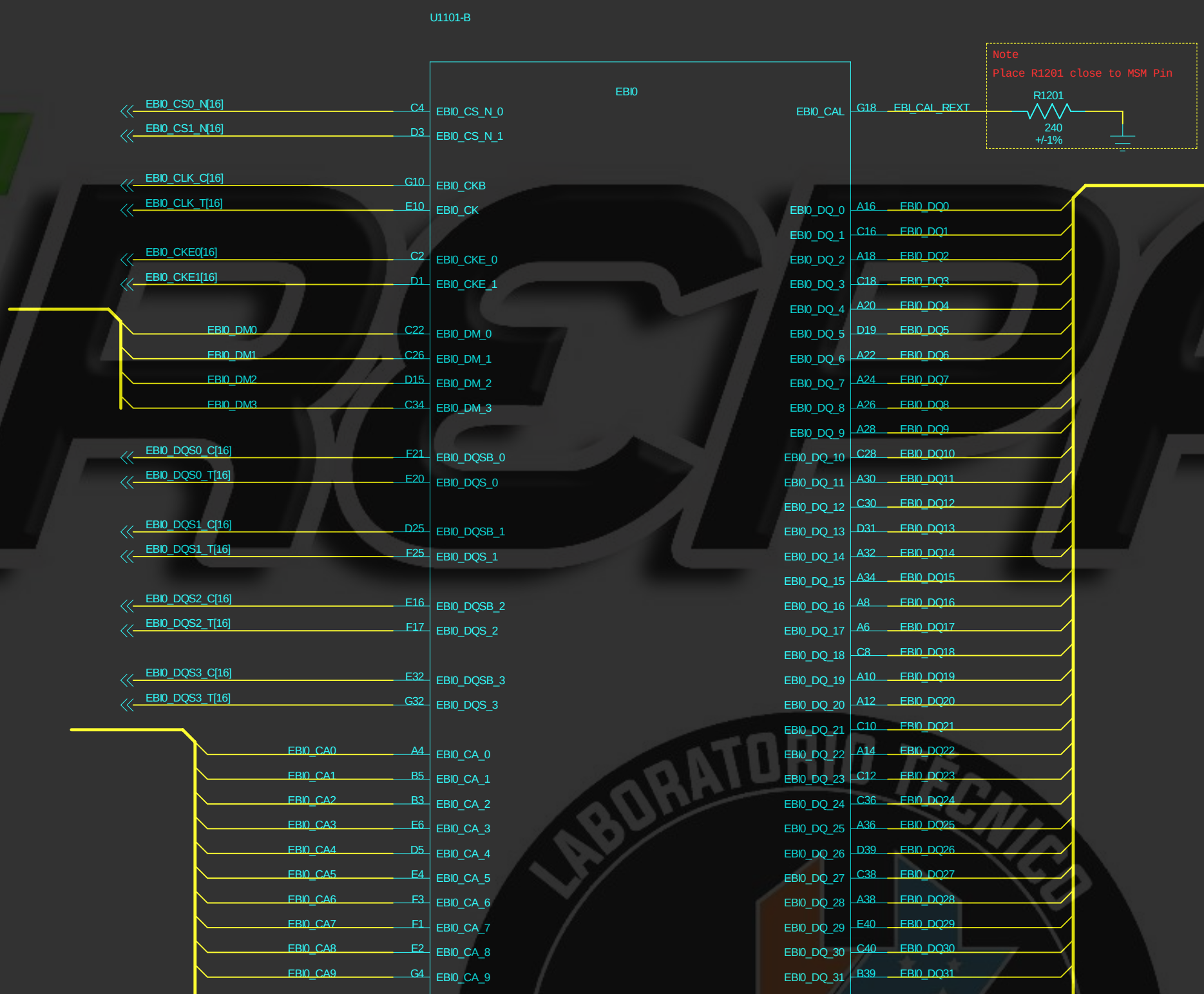
MSM8917 Control

Note: WCSS_XO signal required only for 5GHZ.
When using WCN3615 connect pin BD13 to GND.



DC-DC mode “C For better power ~3 mA lower

MSM8917 EBI



Title

Sheet

Size

Name

Date

1

Sheet

of

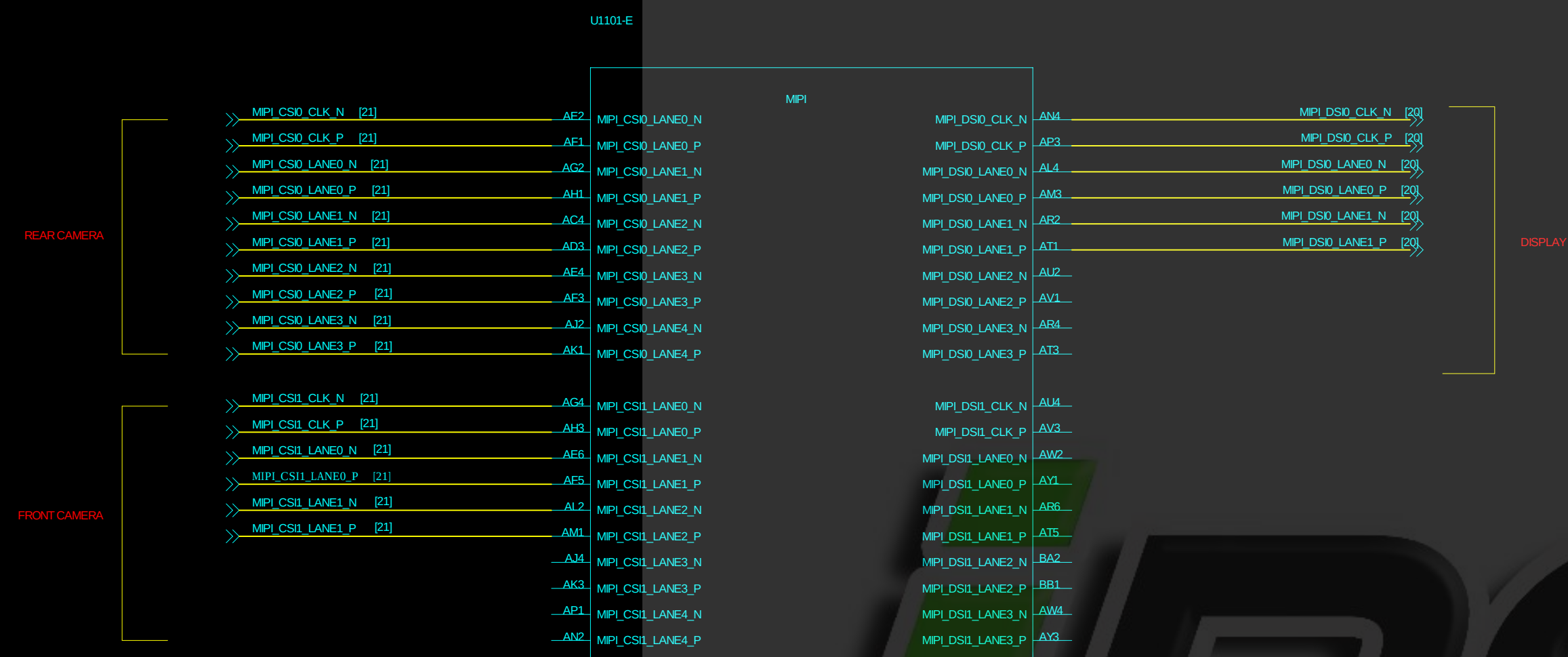
Rev

MSM8917 GPIO



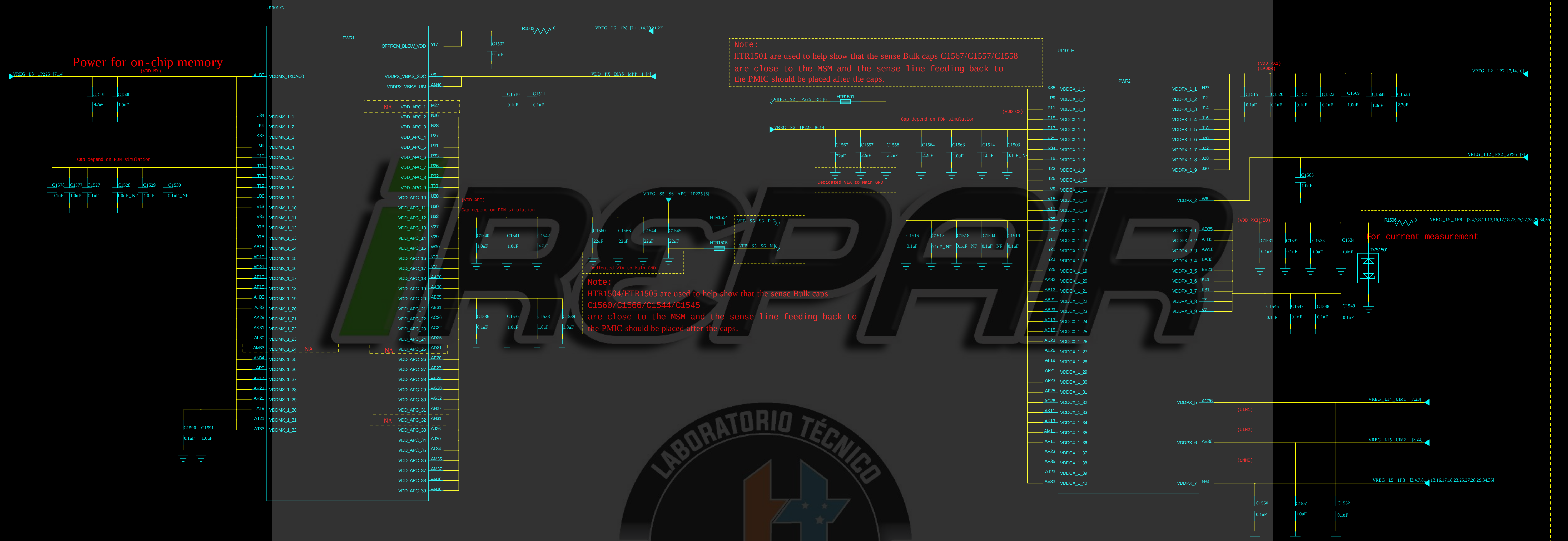
I2C PULL-UP RESISTORS

MSM8917 MIPI

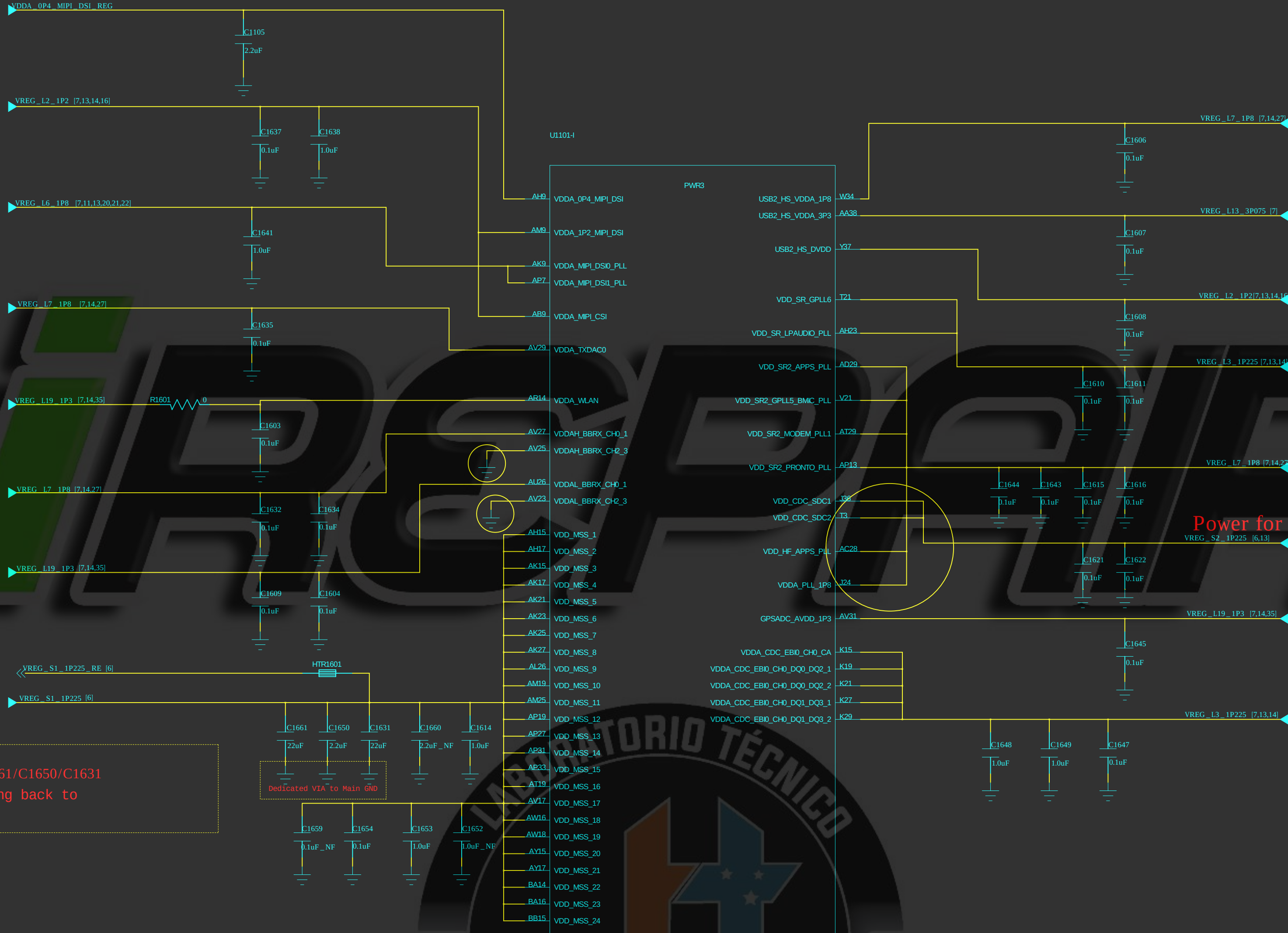


MSM8917 RF Control

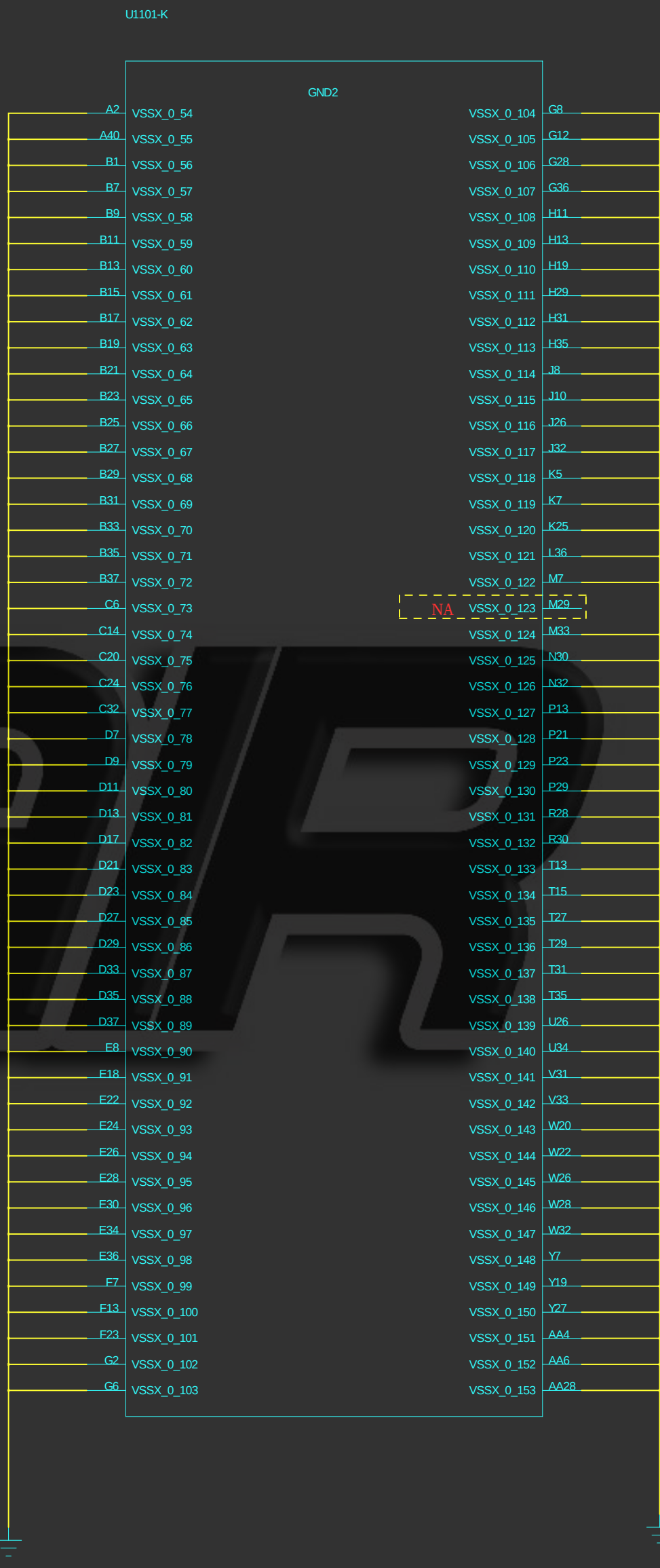
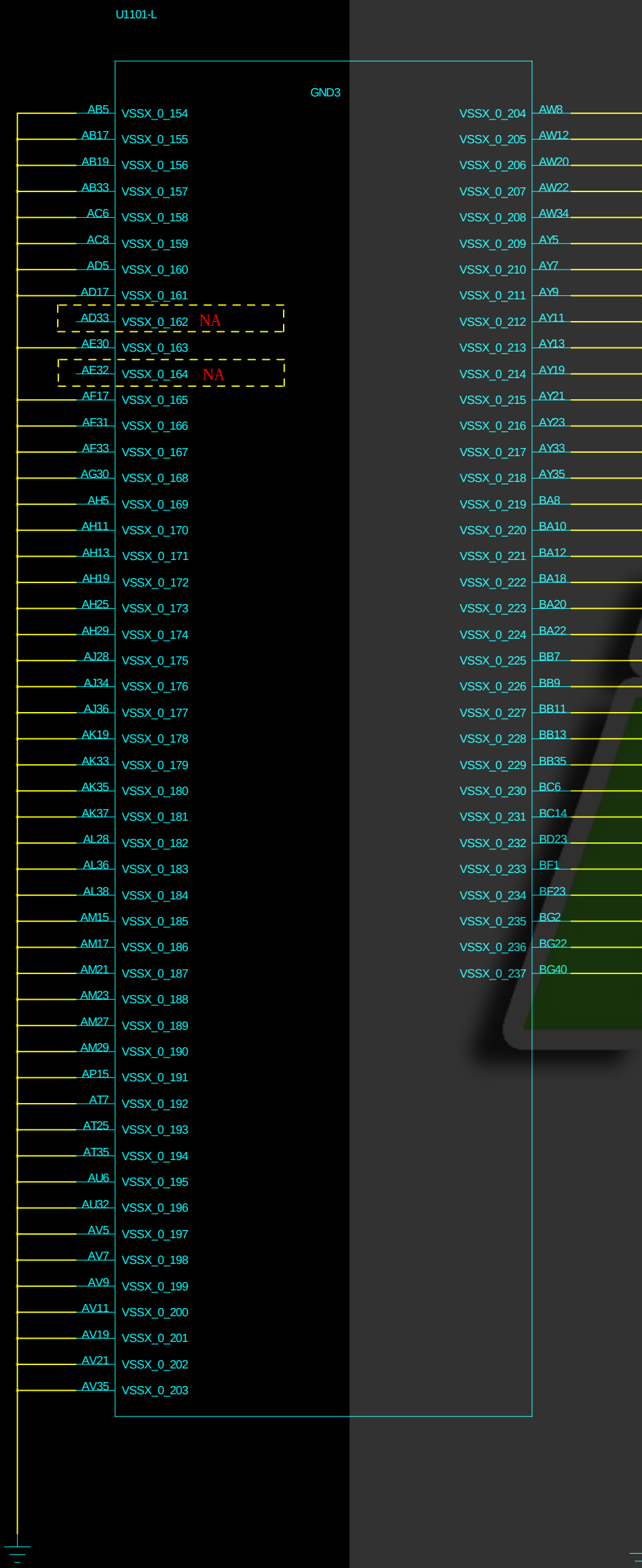
MSM8917 POWER1



MSM8917 POWER2



GND



Title

Sheet

Size

Name

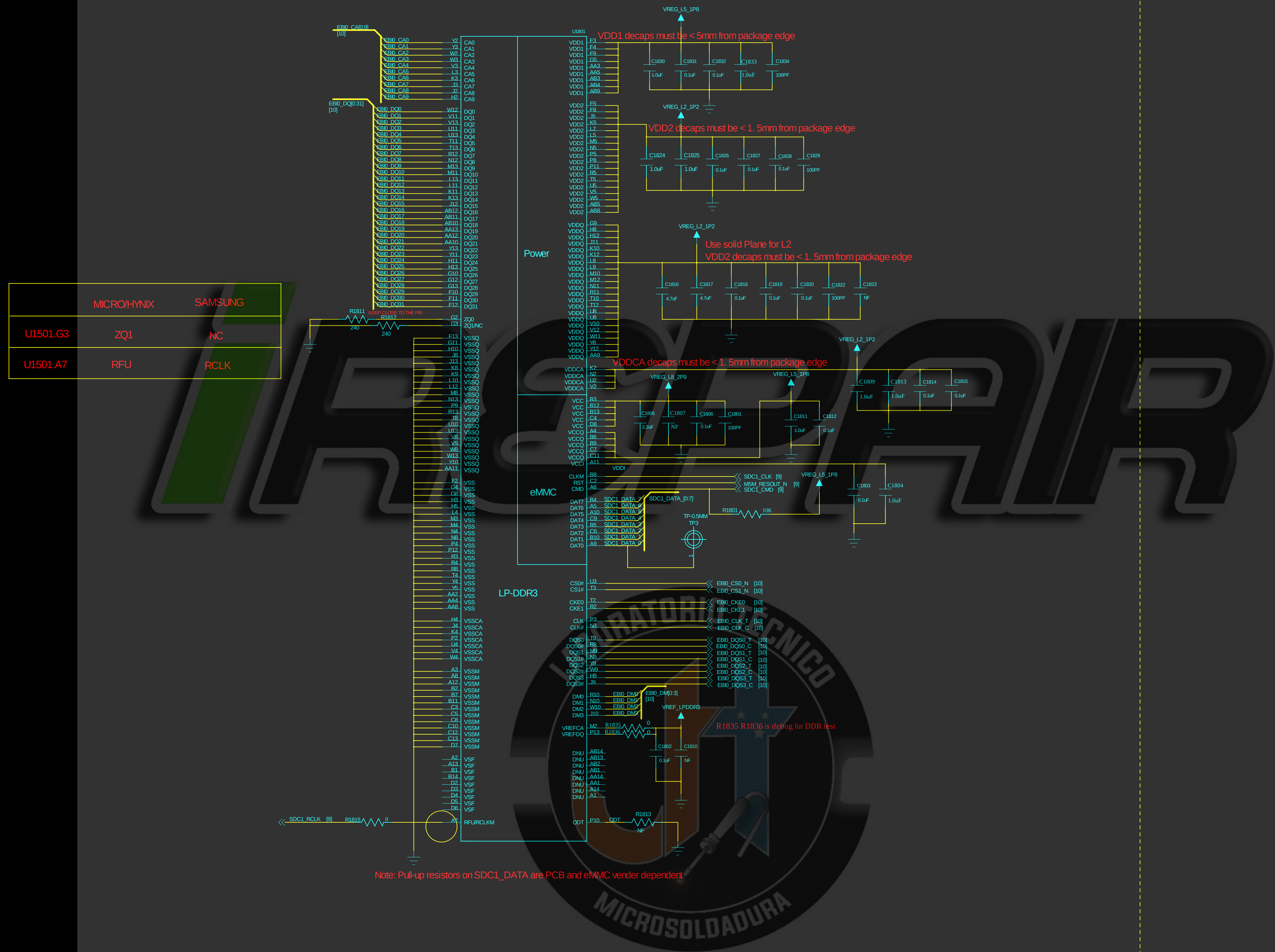
Date

Sheet

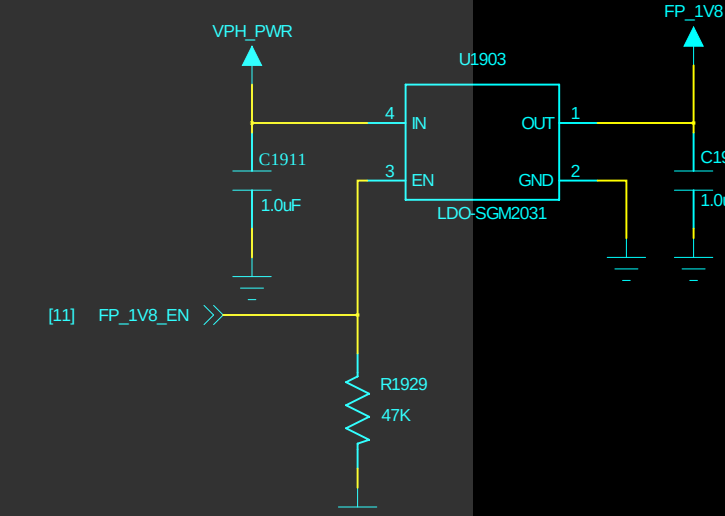
of

Rev

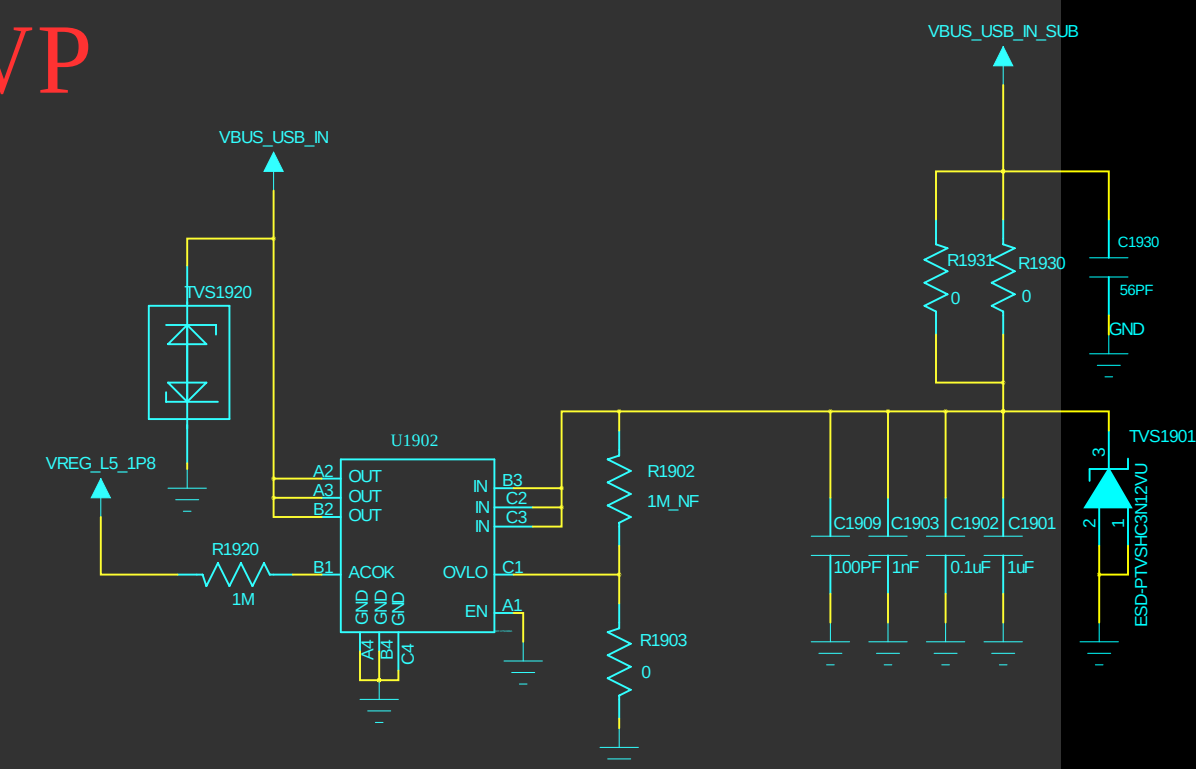
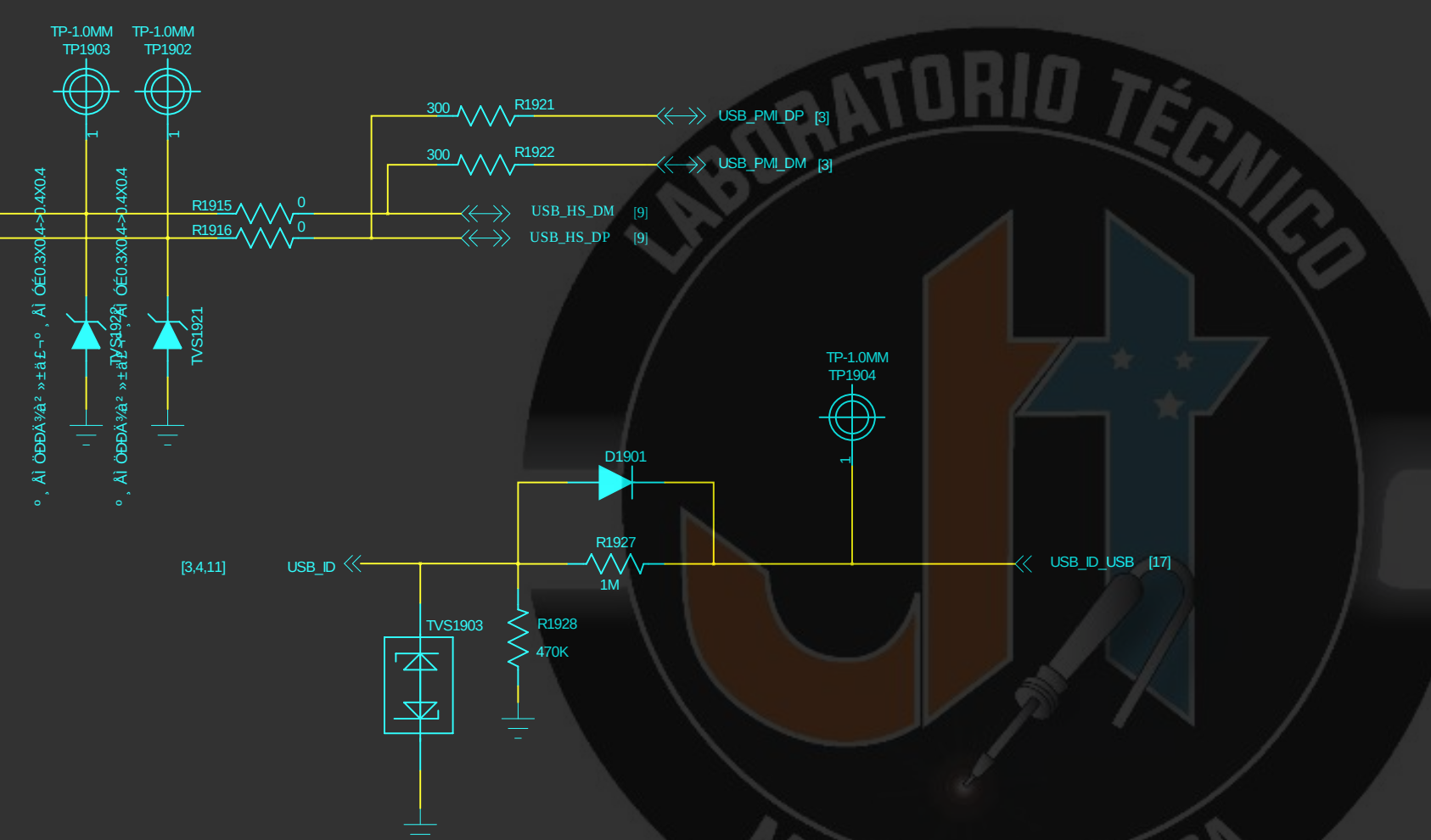
MEMORY:LPDDR3+EMMC



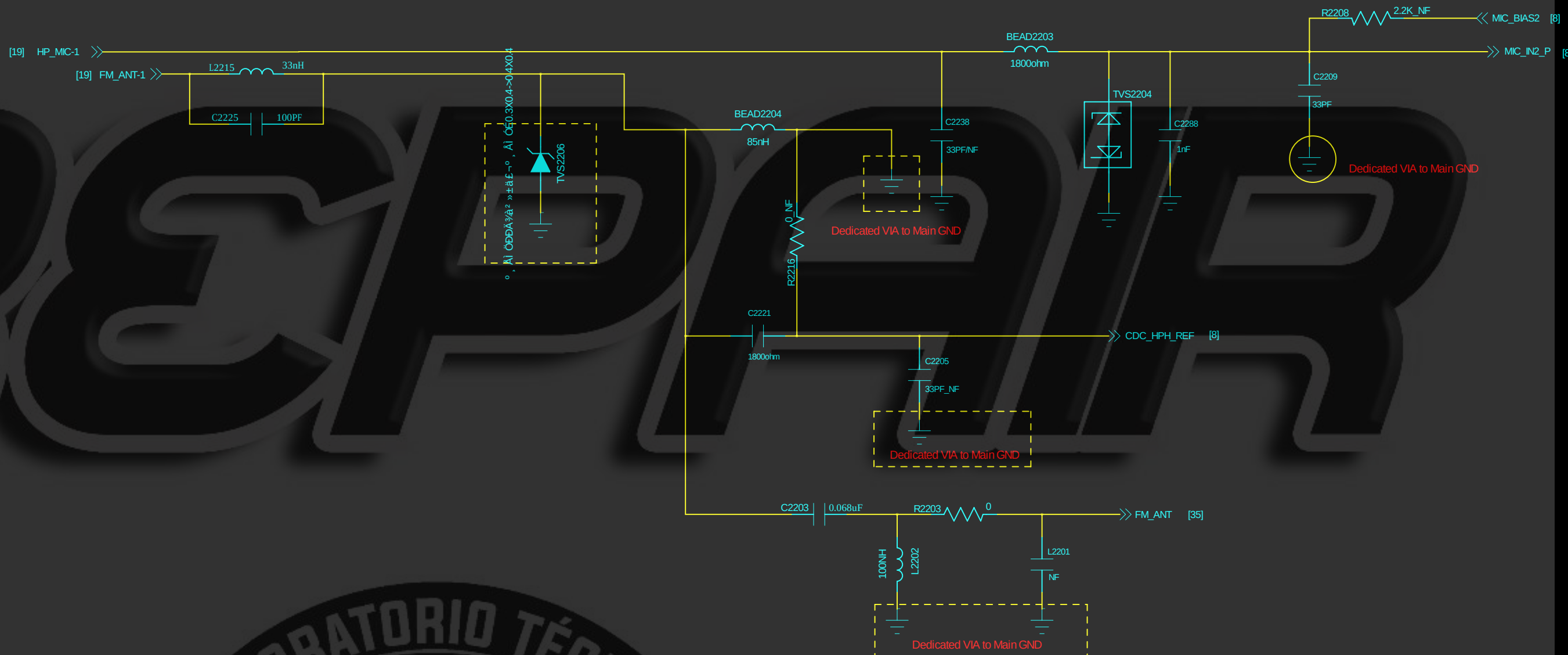
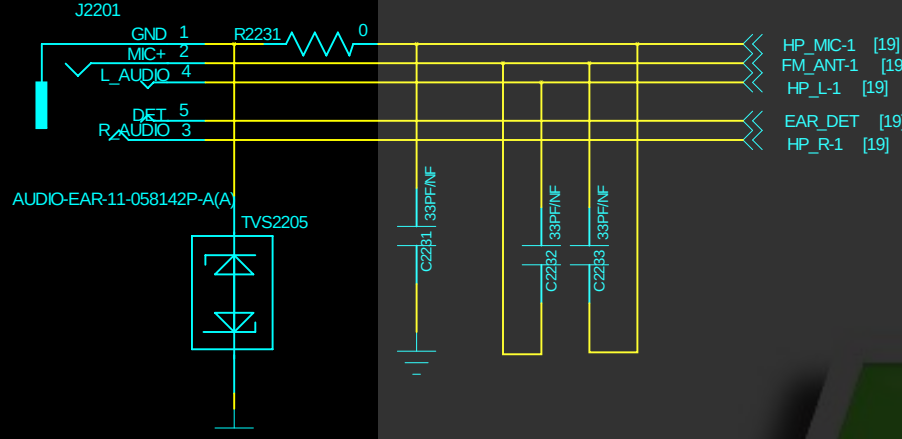
	MICRO/HYNIX	SAMSUNG
U1501.G3	ZQ1	NC
U1501.A7	RFU	RCLK

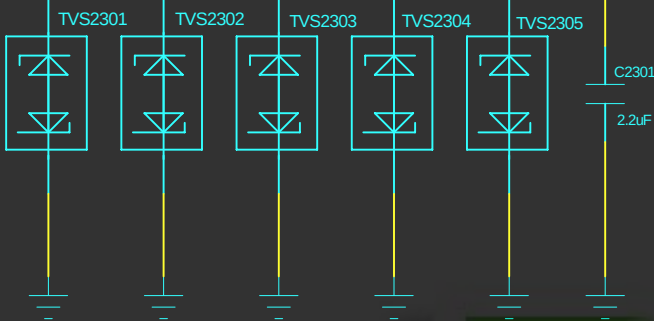


OVP



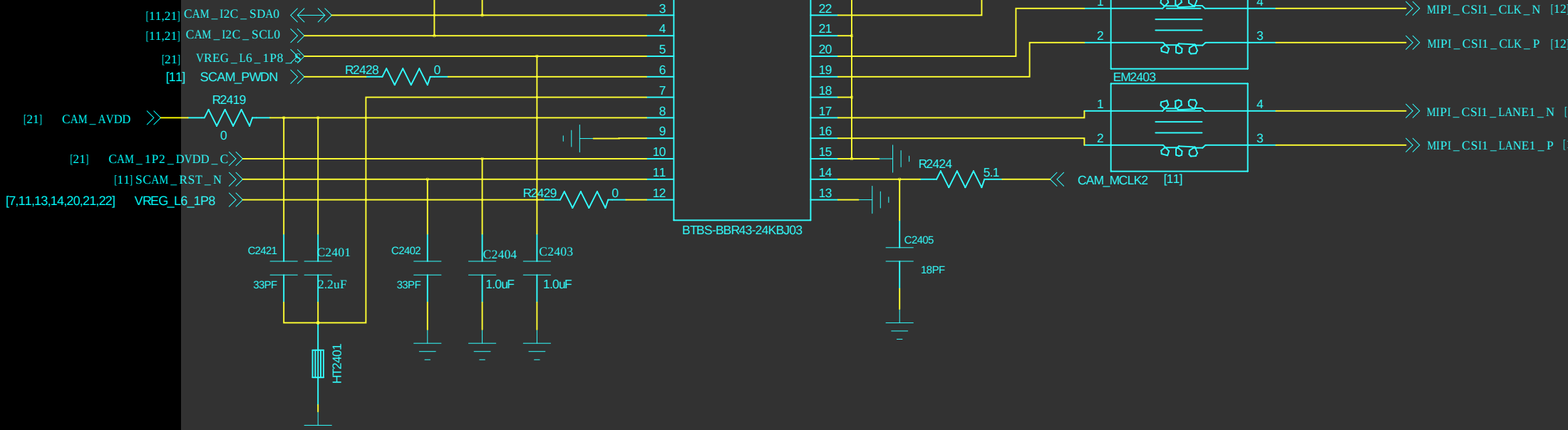
EARPHONE





LCD_VIO28

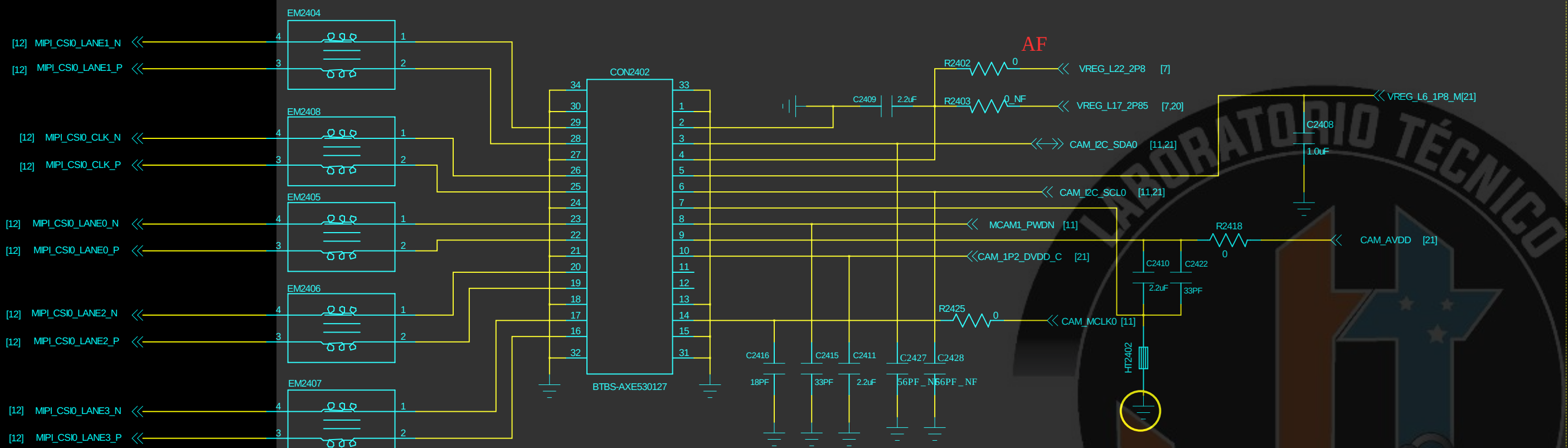
Front Camera



PIN	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
	GND	GND	SDA	SCL	DOVDD	CAM_PON (NC)	AGND	AVDD	GND	DOVDD	CAM_RST	NC	GND	CMMCLK	GND	RDP1_A	RDN1_A	GND	RCP_A	RCN_A	GND	RDP0_A	RDN0_A	GND

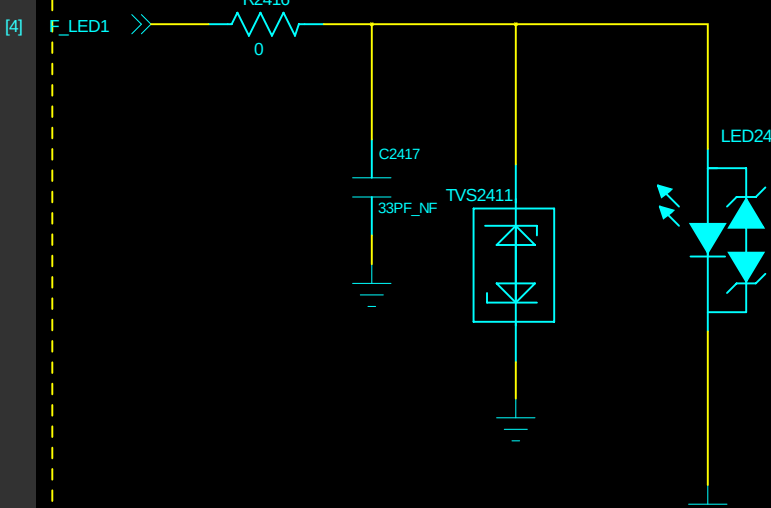
Main Camera / Sub Camera share power domain design
should double check the voltage level is compatible

Main Camera

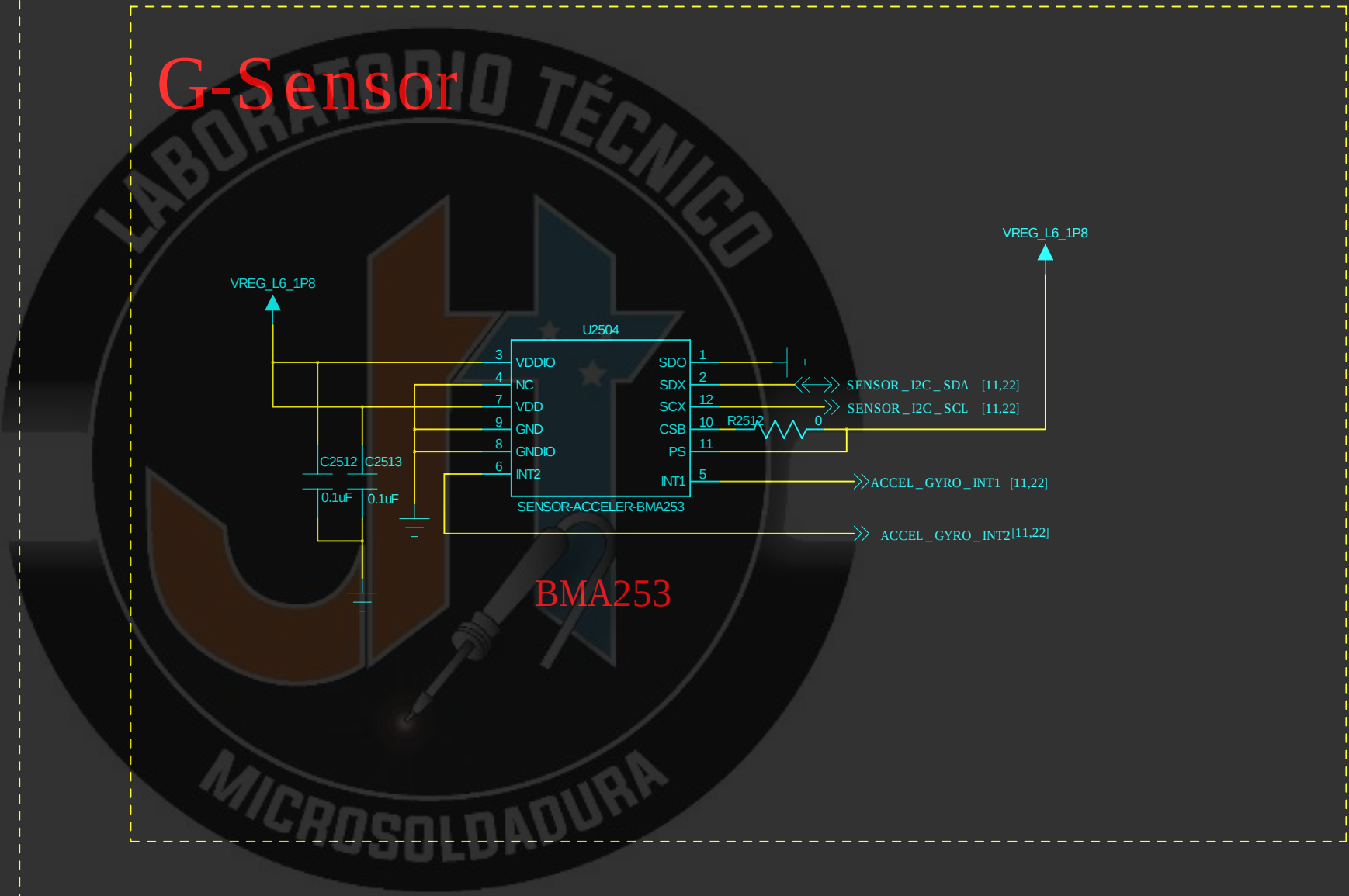
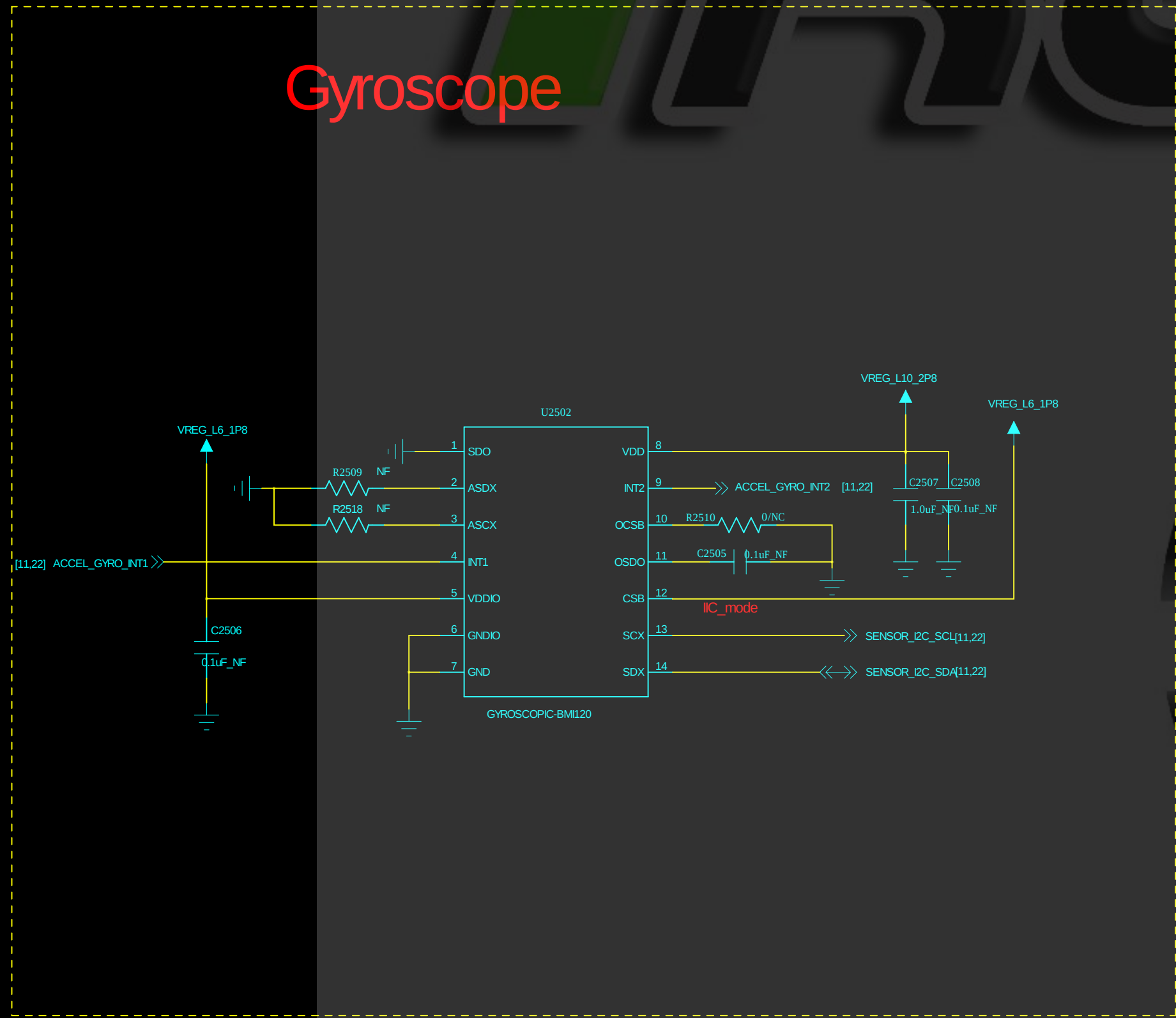
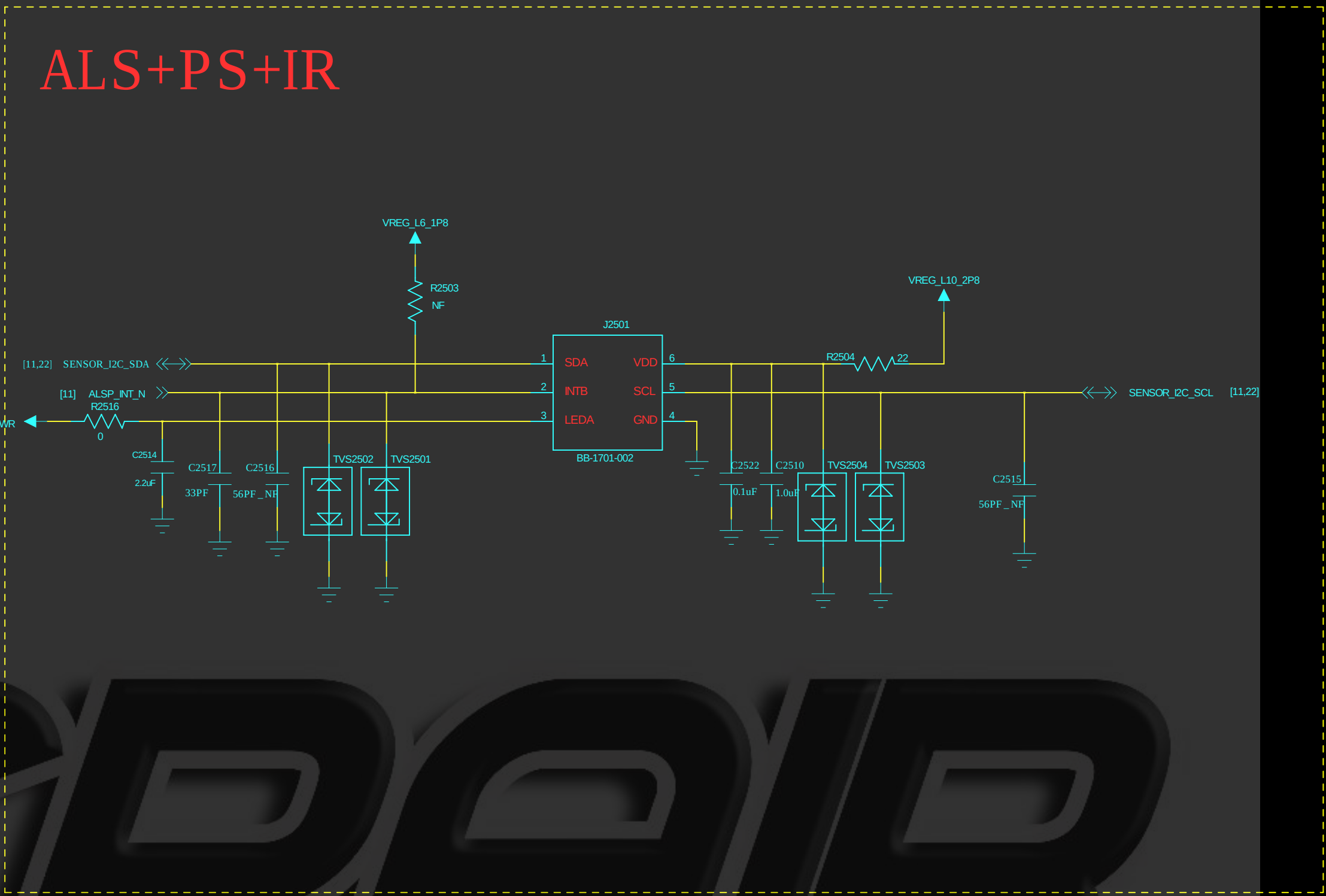
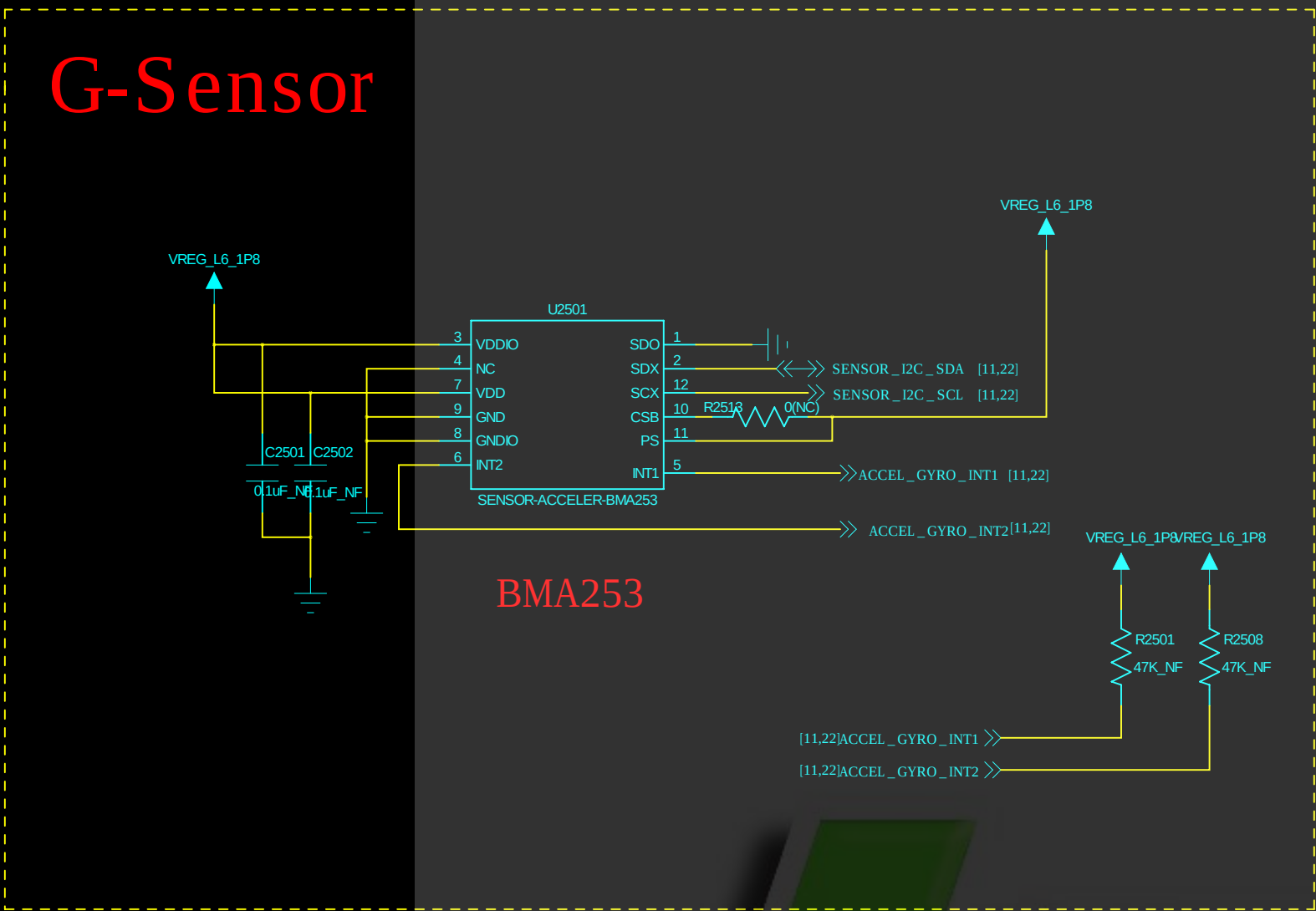


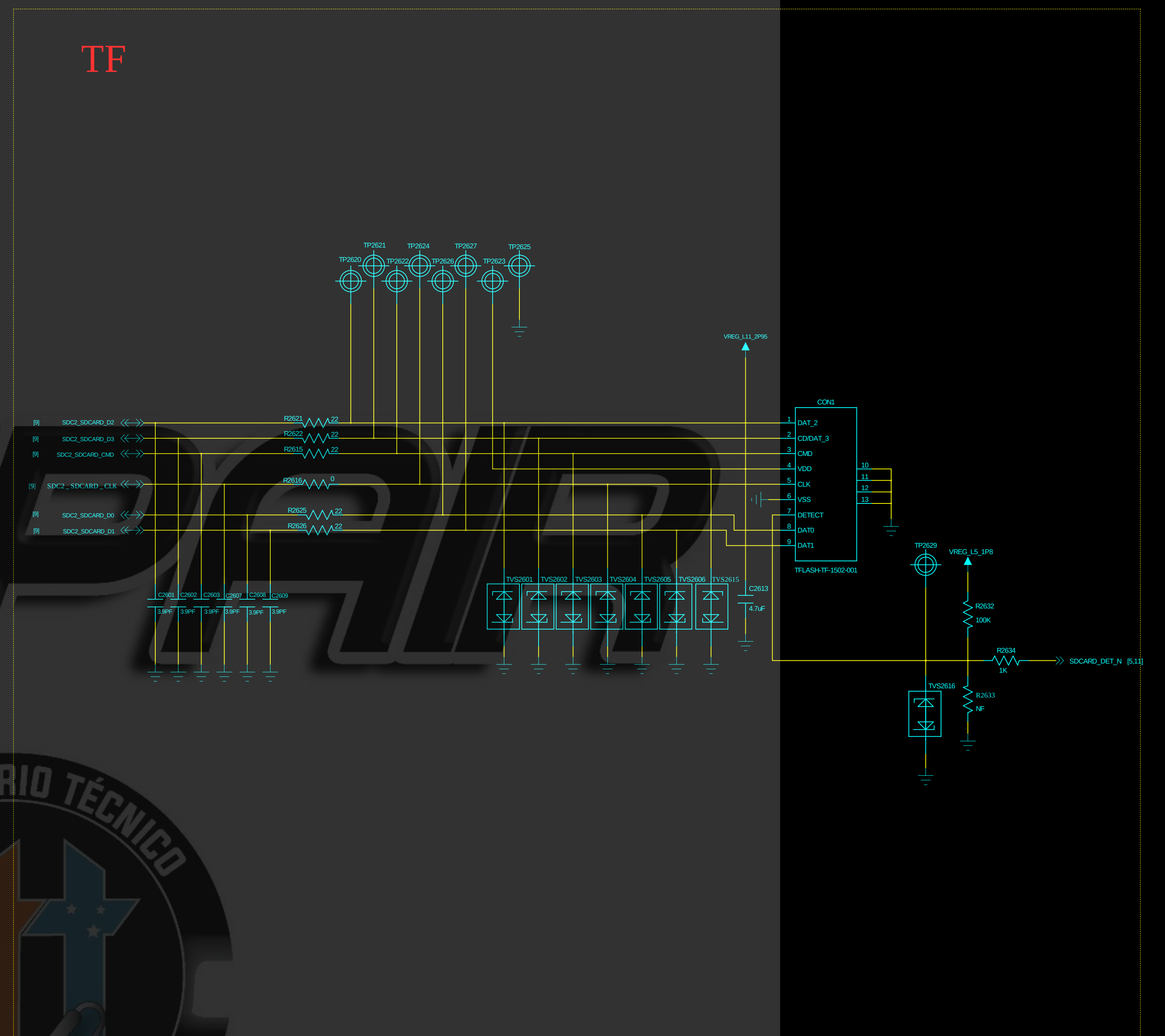
Front Flash LED

Main Flash LED



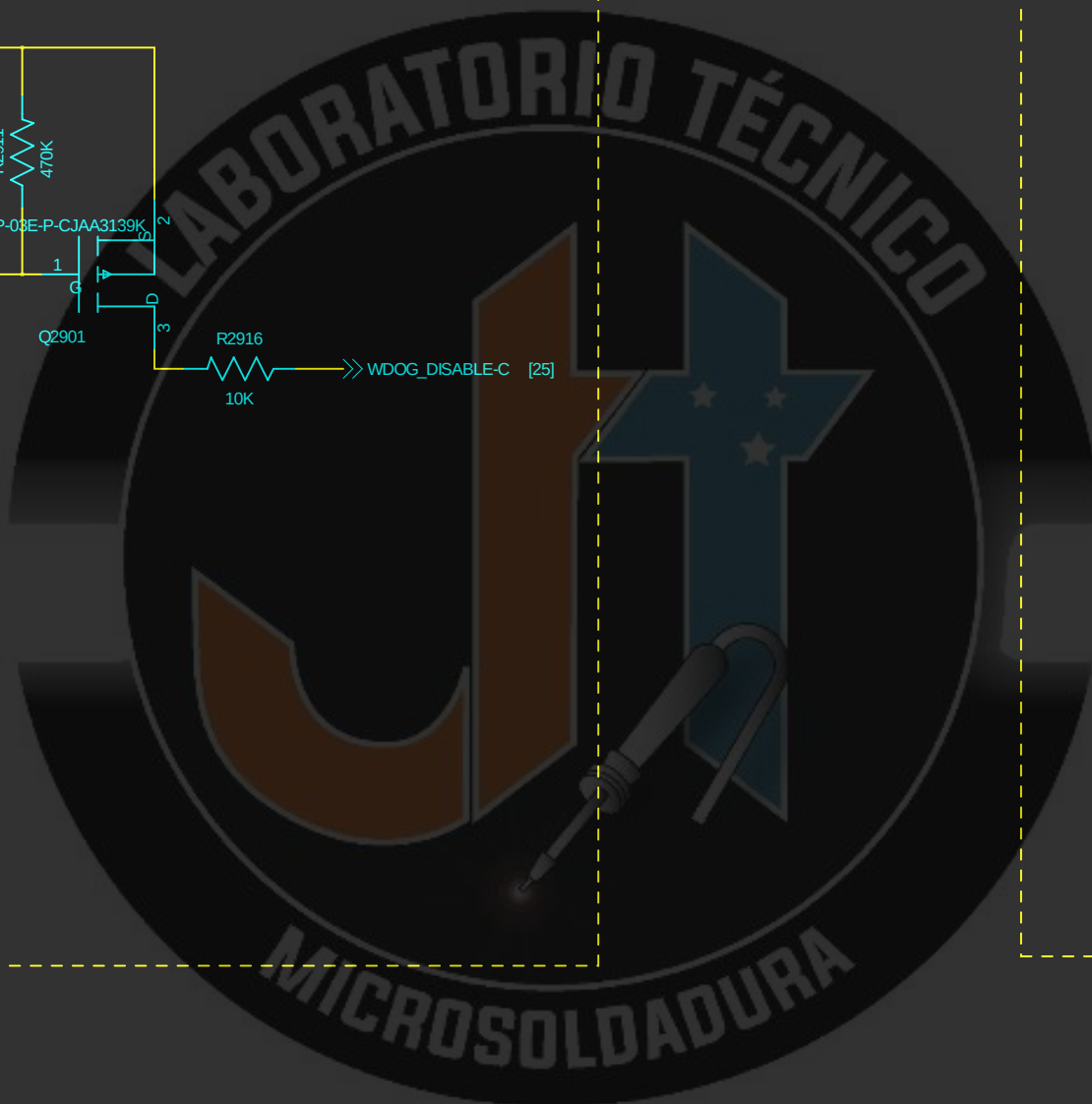
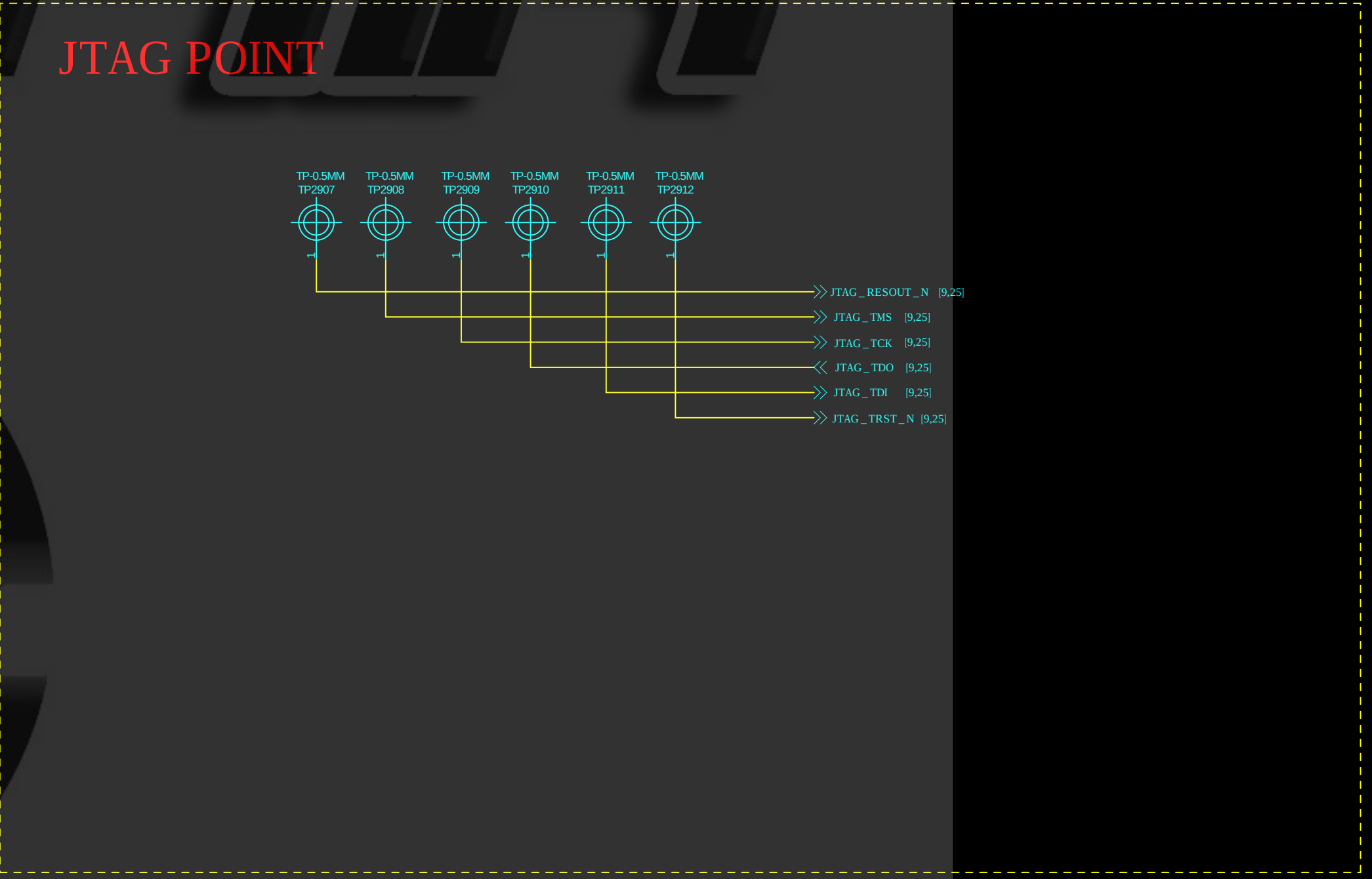
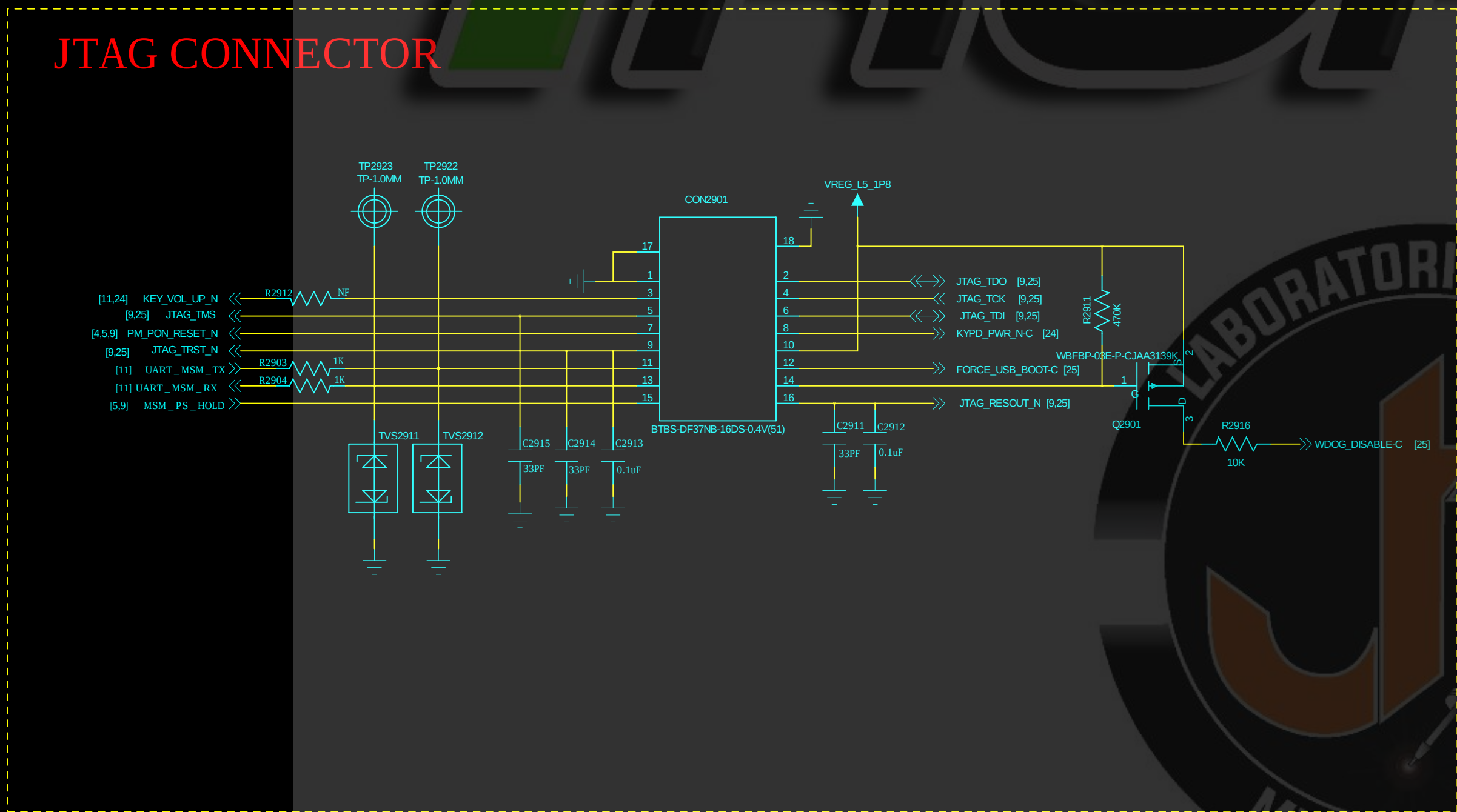
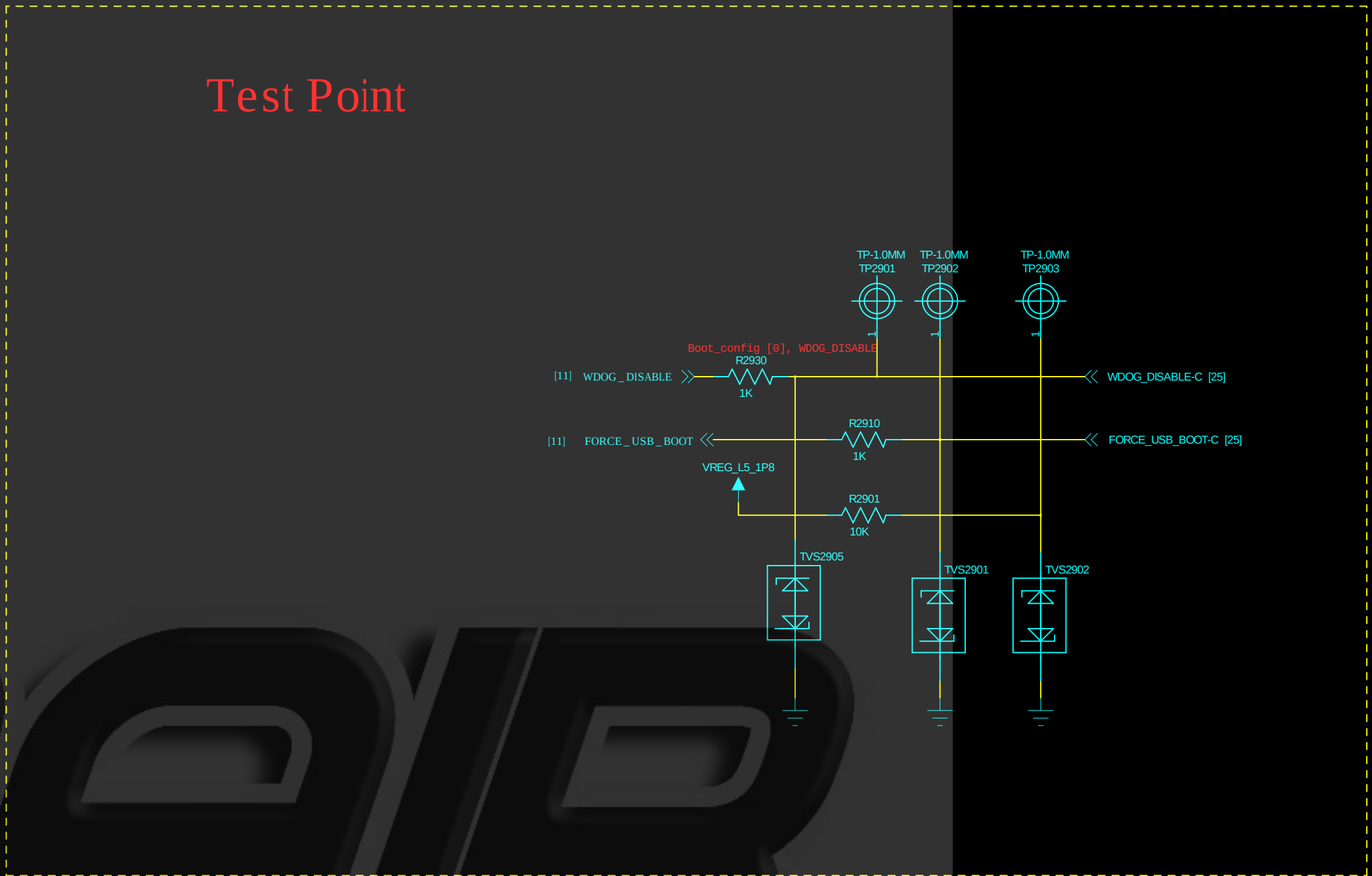
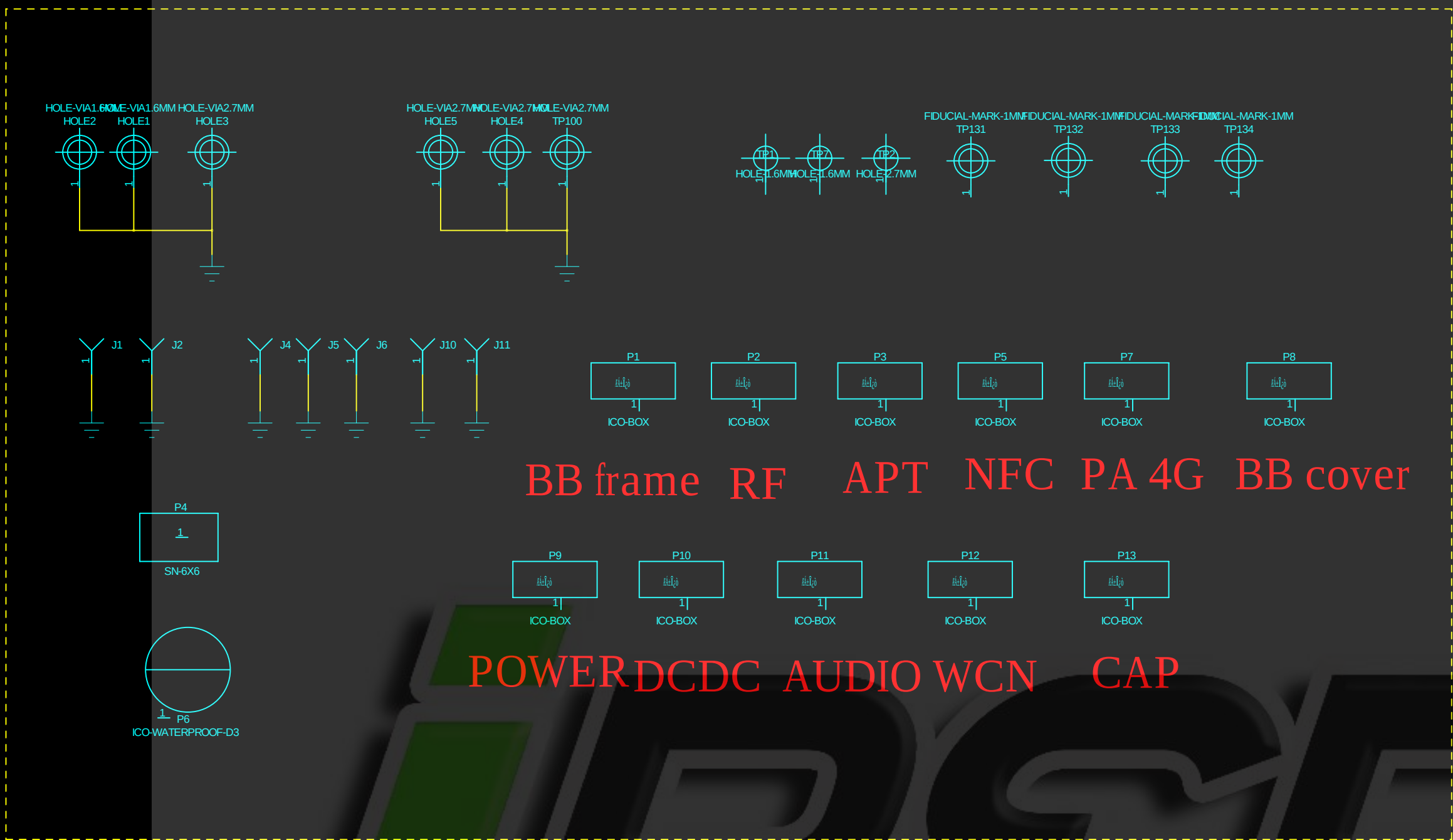
PIN	
1	DGND
2	AF—GND
3	SDA
4	AF—VCC
5	DOVDD
6	SCVDD
7	AGND
8	PWDN
9	AVDD
10	DVDD
11	NC
12	NC
13	DGND
14	XCLK
15	DGND
16	MDP3
17	MDN3
18	DGND
19	MDP2
20	MDN2
21	DGND
22	MDP0
23	MDN0
24	DGND
25	MCP
26	MCN
27	DGND
28	MDP1
29	MDN1
30	DGND





KEYPAD

INDICATOR LED

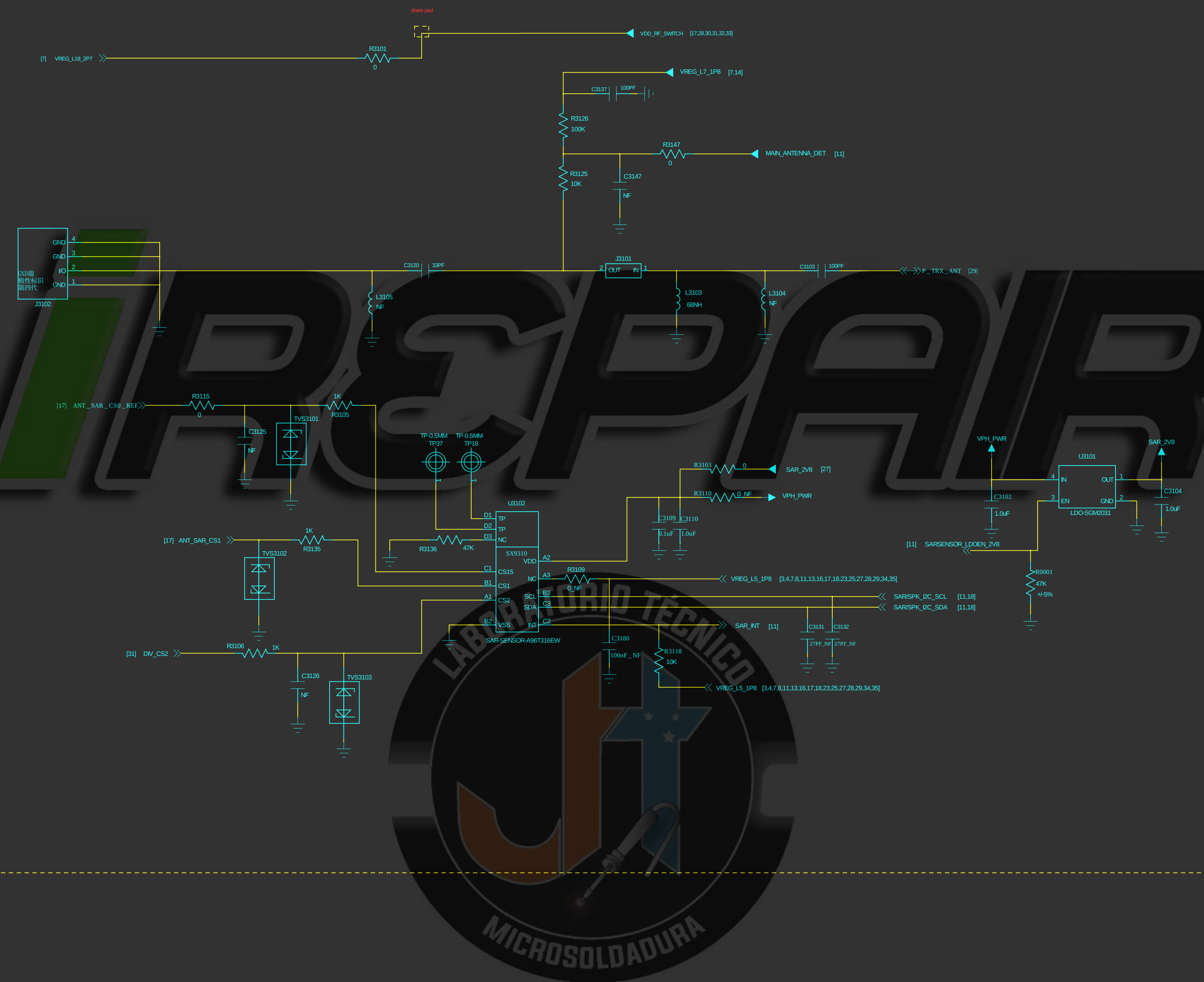


WTR 2965

WTR2965 Tx port mapping						
RF port	GSM	CDMA	WCDMA	LTE-FDD	LTE-TDD	TD-SCDMA
Tx_DA1	850, 900, 1800, 1900	0, 10	8, 11	8, 11, 12, 13, 17, 21, 28	39	34, 39
Tx_DA2	850, 900	0, 10	5, 6, 8, 19	5, 6, 8, 16, 19, 20, 26, 27, 28		
Tx_DA3				7, 30	38, 40, 41	
Tx_DA4	1800, 1900	1, 6, 14, 15	1, 2, 3, 4, 9, 10, 11, 25	1, 2, 3, 4, 9, 10, 11, 21, 25, 66	34, 39	34, 39

Rf port mapping							
WTR2965							
	CDMA	GSM	WCDMA	LTE-FDD	TD-SCDMA	LTE-TDD	Remarks
Prfx_LB1	BC0,10	850,900	5, 6, 8, 19	5, 6, 8, 18, 19, 26, 27			SAWless and SAW mode is not supported at the same time
Prfx_LB2		850 SAWless, 900 SAWless		12, 13, 17, 20, 28			
Prfx_LB3	BC0,10	850, 900	5, 8	5, 6, 8, 13, 18, 19, 20, 26, 27, 28, 29			
Prfx_MB1	BC15 BC1,14	1800 1900	2, 4, 10, 11, 25	2, 4, 10, 11, 21, 25, 32	34 SAW/SAWless 39 SAW/SAWless	34 SAW/SAWless 39 SAW/SAWless	SAWless and SAW mode is not supported at the same time
Prfx_MB2	BC1,14 BC6, 15	1800, 1900	2, 3, 9, 25	2, 3, 9, 25	34, 39	34, 39	
Prfx_MB3		1800	1, 3, 4, 9, 10	1, 3, 4, 9, 10, 66			
Prfx_HB1				30		38, 40, 41	
Prfx_HB2				7		38, 41	
	CDMA	GSM	WCDMA	LTE-FDD	TD-SCDMA	LTE-TDD	
DRx_LB1	BC0, 10	850, 900	5, 6, 8, 19	5, 6, 8, 18, 19, 26, 27			
DRx_LB2				12, 13, 17, 20, 28			
DRx_LB3	BC0, 10	850, 900	5, 8	5, 6, 8, 13, 18, 19, 20, 26, 27, 28, 29			
DRx_MB1	BC15, BC1,14	1900	2, 4, 10, 11, 25	2, 4, 10, 11, 21, 25, 32	34, 39	34, 39	
DRx_MB2	BC1, 14	1800, 1900	2, 3, 9, 25	2, 3, 9, 25	34, 39	34, 39	
DRx_MB3	BC6, 15	1800	1, 3, 4, 9, 10	1, 3, 4, 9, 10, 66			
DRx_HB1				30		38, 40, 41	
DRx_HB2				7		38, 41	

PRIMARY_ANT



Title	
Sheet	Rev
Size	mm
Date:	Sheet of

3/4G PA

Band7

Band40

Band4

Band3

Band1

Band2

Band5/BC0

Band20/28A

Band8

Band28B

2G PA

BOM PA : RF5212A

GSM_HB

2G3G4G_LB

B5_B8_B20_B28



Title		
Sheet	Item	Rev
1	1	1



Band2



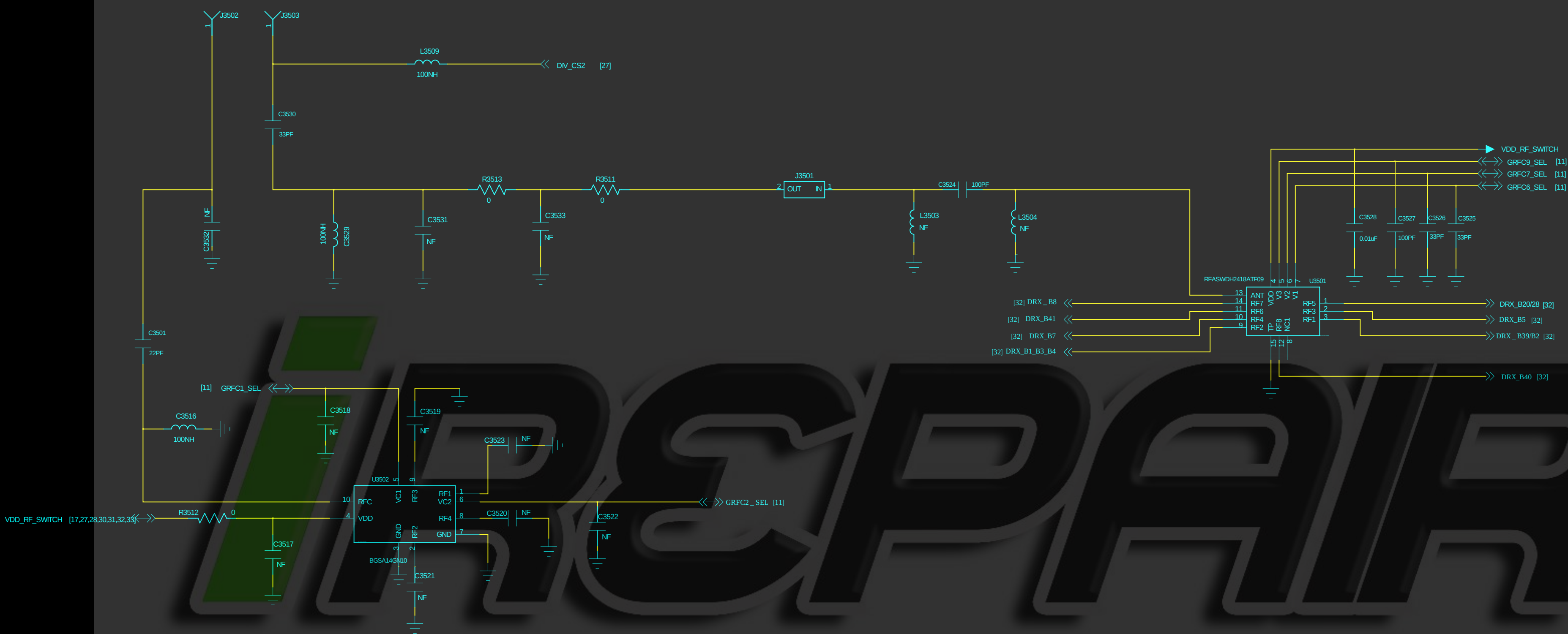
WTR0_PRX_LB1



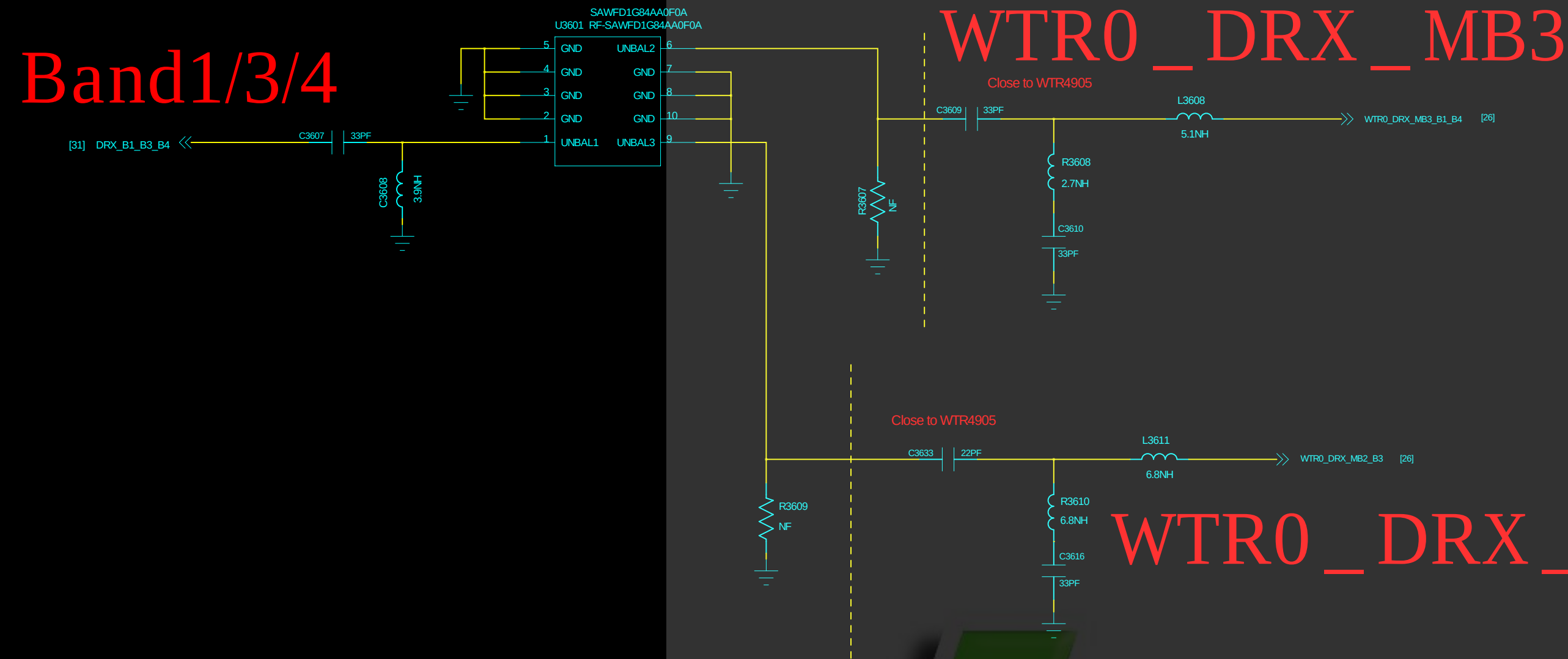
WTR0_PRX_LB3



DRX_ANT



Band1/3/4



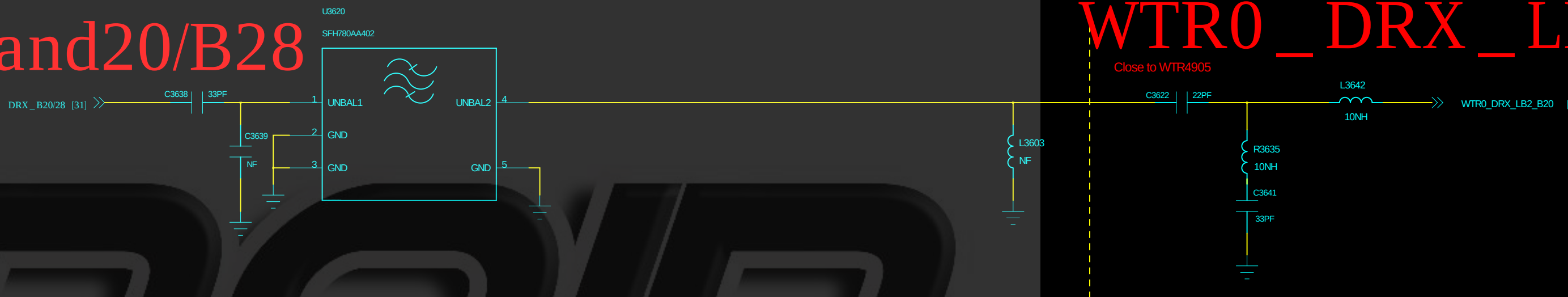
Band5



WTR0_DRX_LB1

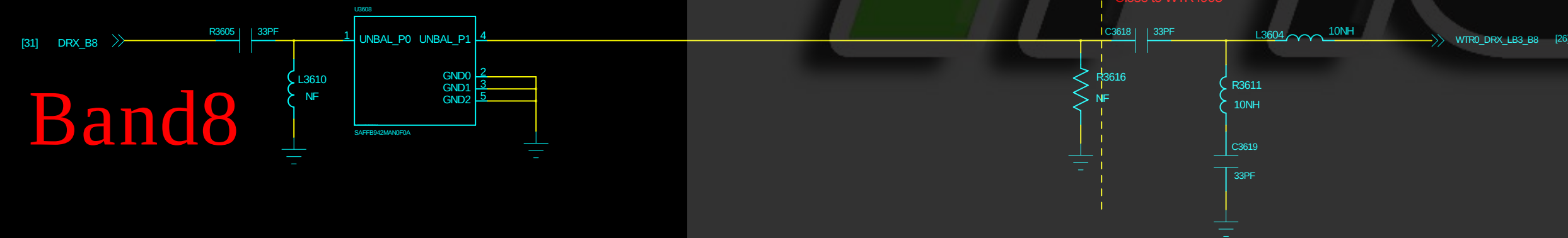
WTR0_DRX_MB2

Band20/B28



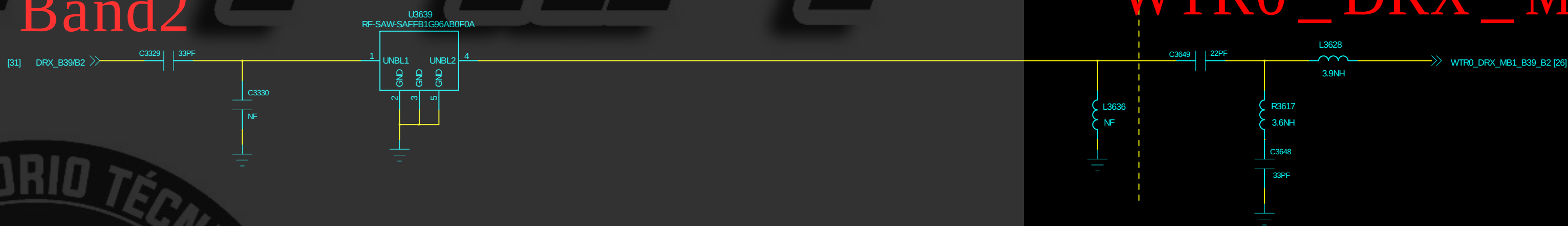
WTR0_DRX_LB2

Band8



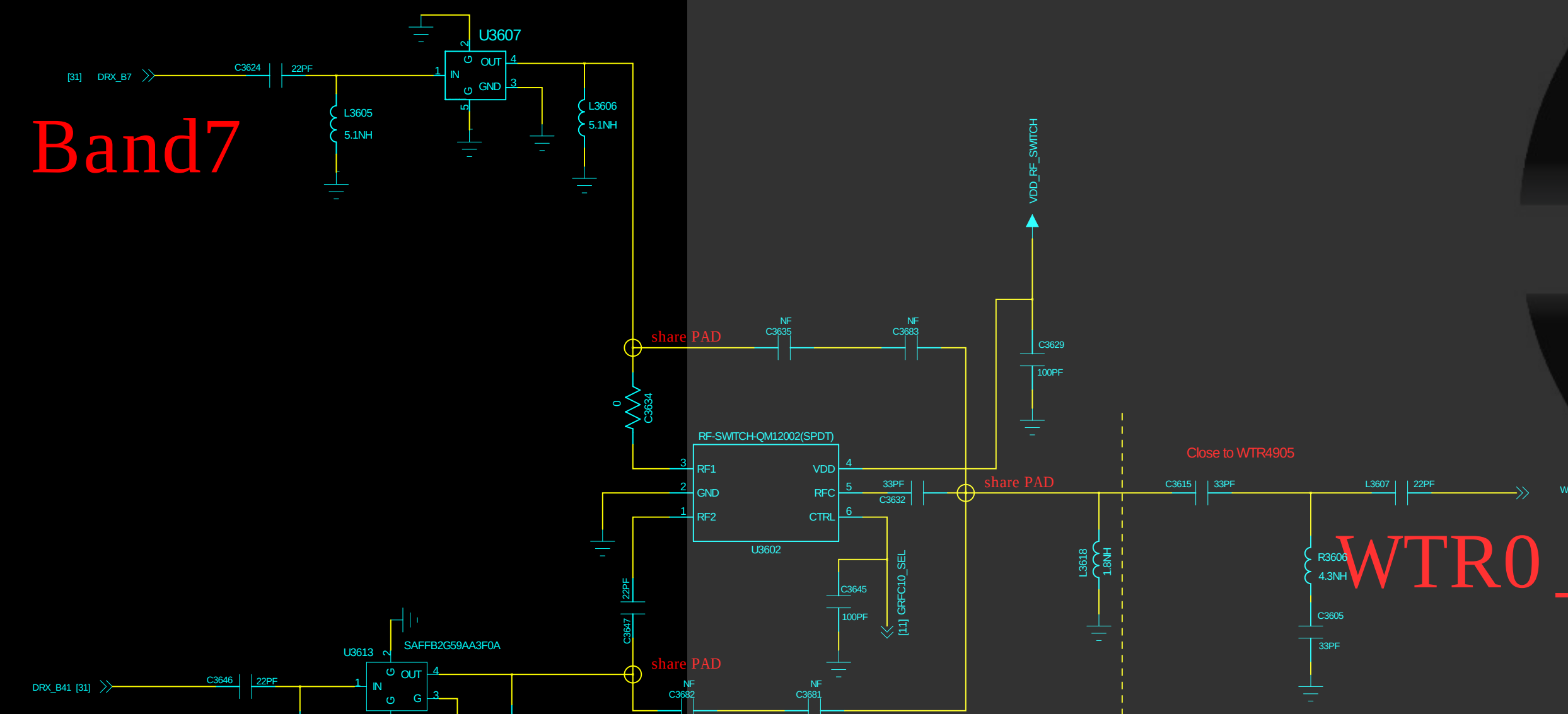
WTR0_DRX_LB3

Band2



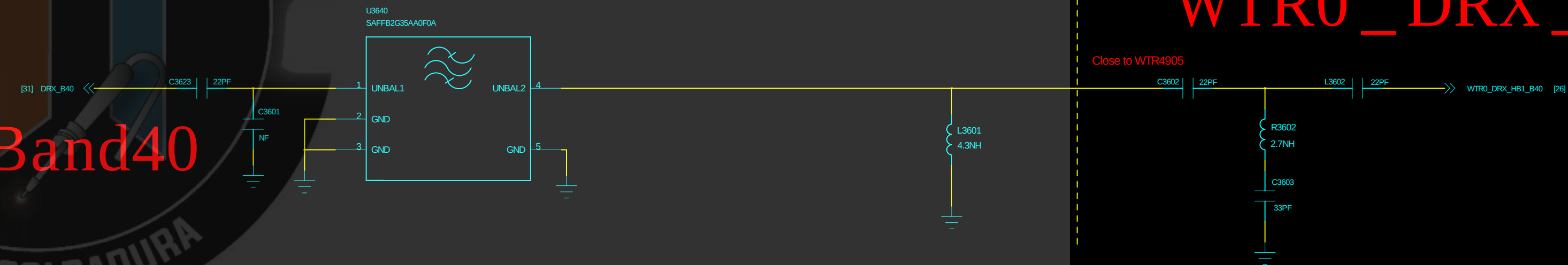
WTR0_DRX_MB1

Band7



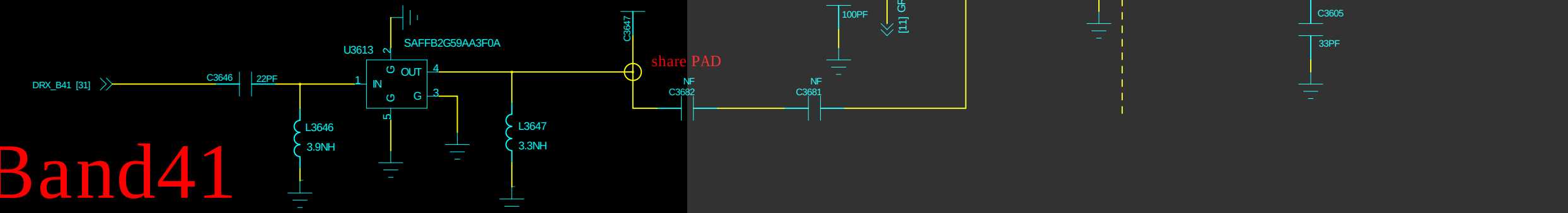
WTR0_DRX_HB2

Band40



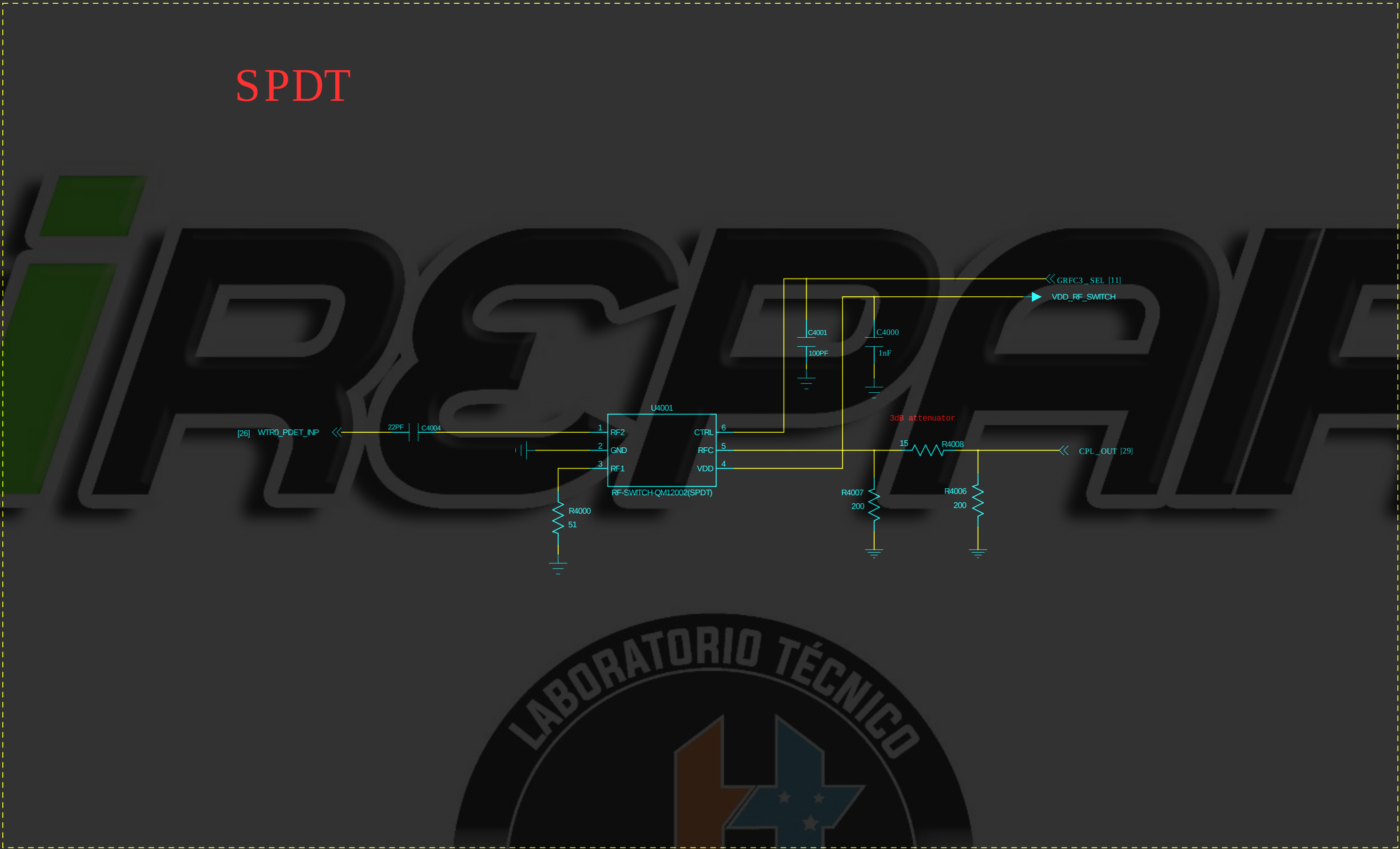
WTR0_DRX_HB1

Band41



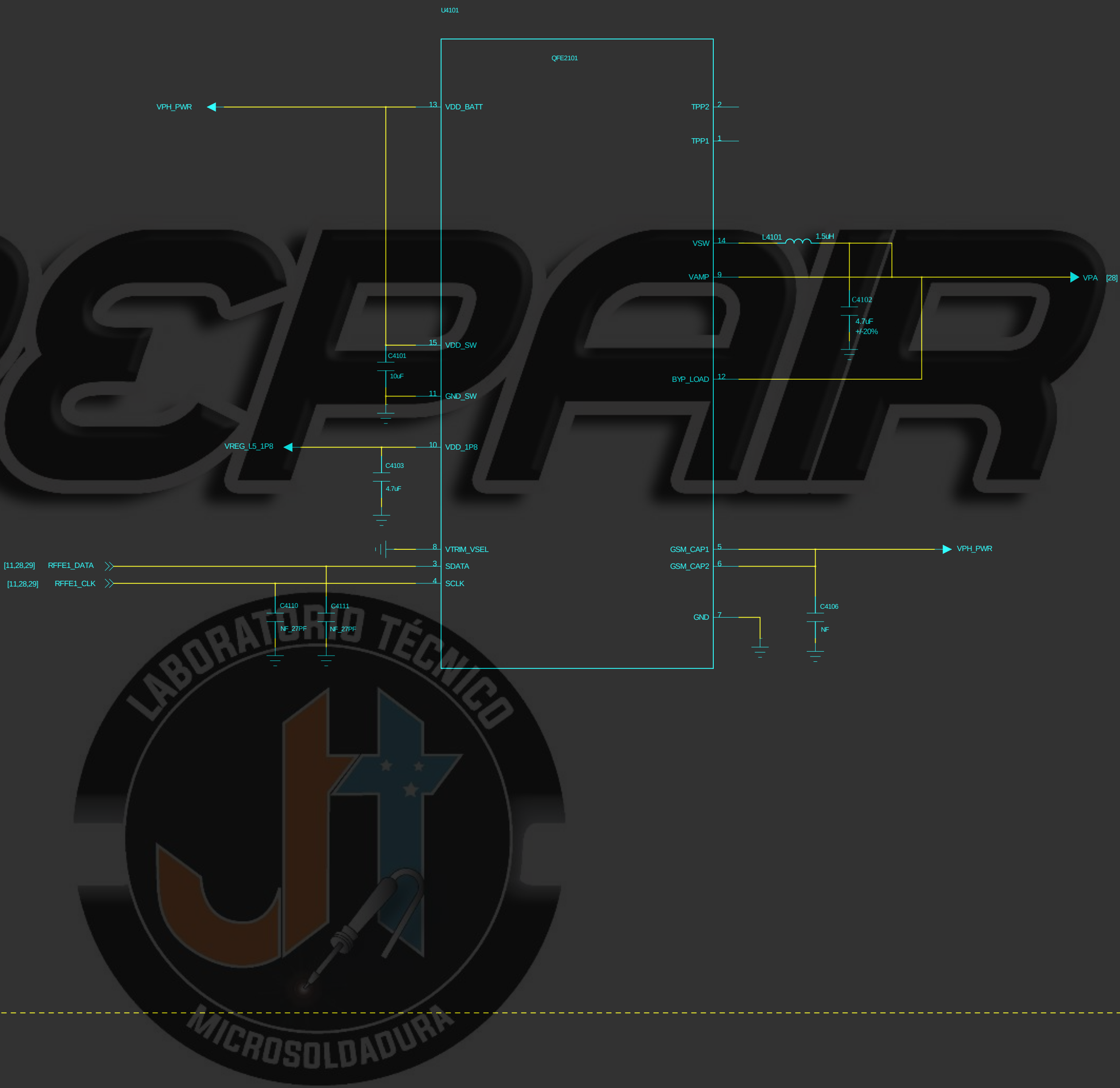
Note:The matching need close to WTR, RX ports have DC at the pin, so it need DC block,

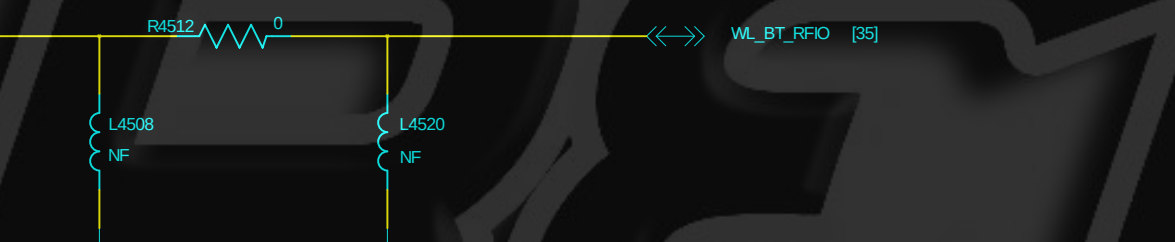
Title		
Sheet	1	1
Rev		



APT

Note: Use a single low impedance power plane/fill VPA for all ET PA VCC1/VCC2.
Refer to 80-NA681-91 rev.B or later revisions for more layout details.





GPS

